Project Report

infrared.city-1: Tree detection

Moritz Hörmansdorfer (h12417021)
Sara Klasová (h12122714)
Dominik Oberhumer (h01025454)
Nina Salnikow (h12109432)

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Abstract

In this project, we developed a machine learning model to detect trees in urban areas using free *Sentinel-2* satellite images. Knowing exactly where trees are is crucial for our partner *infrared.city* to simulate urban temperatures during the summer. We combined public “tree register” datasets from different European cities with *OpenStreetMap* data to train a specialized dual-head *U-Net* model. Because the public city data only lists trees on streets and in parks, but completely misses trees in private gardens, our formal evaluation scores initially appeared rather low. However, visual checks showed that the model actually works very well and successfully learned to find these missing private trees. By using images from three different seasons (spring, summer, and autumn) and a temporal attention mechanism, the model also learned to handle different climates and changing leaf colors. Ultimately, we built a working, scalable prototype that can be used in the future to improve temperature and microclimate simulations for cities.

Contents

1 Introduction

This section sets the context for the research by outlining the underlying motivation, the proposed methodological approach, and the boundaries of the study. It guides the reader from the broader societal challenge of urban climate adaptation to the specific technical implementation of tree detection using satellite imagery.

1.1 Motivation and Project Overview

As urban areas expand and global temperatures rise due to climate change, being able to understand how trees can help cool cities has become increasingly important [1]. Furthermore, trees reduce the risk of flooding in cities and promote health and wellbeing [2]. Our project aims to take the first step in the direction of better tree coverage in urban areas by creating a model that is able to predict precise tree locations using satellite imagery of densely populated areas.

This research builds upon the foundation laid by previous projects conducted in collaboration with the partner company and environmental simulation platform *infrared.city*. A recent flood risk analysis developed for the platform highlighted that surface impermeability is a dominant factor in urban flooding [3]. However, the authors noted that the lack of detailed, highly localized environmental data limits the precision of urban risk analyses. Concurrently, a second project successfully mapped general urban green spaces using *Random Forest* models on *Sentinel-2* data [4]. While effective for broad classification, that project grouped all vegetation into a single category and reported that initial attempts to use *U-Net* architectures for detailed spatial analysis failed to yield good results. The authors explicitly recommended future work to revisit U-Net architectures and to distinguish vegetation types more effectively.

Recently, there has been a rapidly growing body of research on urban tree detection using deep learning models. One of the most recent papers is by Gong et al. [5], who use satellite data not only to detect trees but also to estimate their size and density. Our project tries to build on both this literature and the identified shortcomings of the previous *infrared.city* projects. By integrating multi-season imagery and temporal attention algorithms, we aim to overcome the previous limitations of the U-Net model and achieve a more granular, object-specific classification than general green spaces.

1.2 Proposed Approach and Data

To achieve reliable urban tree detection, our approach combines multi-spectral satellite imagery with a deep learning architecture optimized for image segmentation.

We utilize publicly available Sentinel-2 satellite images, providing a 10-meter spatial resolution across four color channels: red, green, blue, and near-infrared. These images are retrieved via the API of the *Copernicus Programme* and are provided by the European Space Agency [6]. A key technical focus of our data strategy is the inclusion of imagery from different seasons to account for the significant impact of seasonality on tree appearance. To provide ground truth for training and testing, we integrate publicly available "tree register" datasets, which contain exact locations for registered trees across multiple European cities [7, 8, 9, 10, 11, 12].

Regarding the model, we employ a U-Net architecture, a specialized *Convolutional Neural Network* (CNN) consisting of symmetric contracting and expanding paths [13]. While a U-Net implementation in a previous project did not yield promising results, this paper aims to build upon that foundation as the architecture is considered highly accurate for detecting objects in satellite data [14, 15]. Our model extracts features from seasonal multi-channel imagery through its contracting path and utilizes an expanding path for upsampling to predict precise locations.

To improve upon previous attempts, our implementation incorporates skip connections and a temporal attention algorithm to prioritize more meaningful images within the time series [15]. The model performance is optimized using Binary Cross-Entropy loss, comparing our predictions against the split training and test sets derived from these multi-city "tree registers".

1.3 Scope and Limitations

The model in this paper is trained only on satellite images of urban areas and is therefore only supposed to be used for tree detection within city environments. Additionally, the model focuses on tree detection in the temperate climate zone, as this is the vegetation it is trained to observe. Further analyses on the impact of trees on, for example, heat regulation in cities and flooding prevention are beyond the scope of this project and could be the topic of future research.

Since this paper uses Sentinel-2 satellite images with 10-meter spatial resolution, issues could arise, as the images might not have a high enough resolution to detect trees reliably. Future work could therefore also focus on improving image quality by using higher-resolution satellite data.

1.4 Source Code Availability

The source code, datasets, and scripts used to produce the results in this project are open source and can be found in the following GitHub repository:

`https://github.com/nikmaster/wu_dslab_infrared-city`

2 Related Work

This section provides an overview of the current state of research in urban tree mapping. It explores the transition toward advanced machine learning, discusses the strengths and limitations of high-resolution imagery, and examines the role of multi-spectral and multi-temporal data in scalable vegetation analysis.

2.1 Transition to Deep Learning in Tree Detection

The detection and classification of urban trees using remote sensing imagery is a rapidly evolving field, characterized by a transition from traditional algorithms to advanced machine learning techniques. A comprehensive overview is provided by Zheng et al. [16], who review various methods for individual tree crown detection and delineation from optical imagery, highlighting the growing reliance on deep learning architectures. Building on this methodological shift, Durgut et al. [17] demonstrate the benefits of integrating traditional image processing methods with modern deep learning approaches to further enhance the accuracy of tree crown detection in satellite images.

2.2 High-Resolution Imagery and Its Limitations

A significant portion of recent research focuses on highly detailed classification tasks utilizing very high-resolution (VHR) imagery. For instance, Sicard et al. [18] explore object-based classification techniques to accurately distinguish different urban plant species. Similarly, Wenger et al. [19] focus on improving urban tree species classification by combining VHR satellite data with various machine learning models. While these approaches achieve high precision at the individual tree level, they often depend on commercial or less frequently updated satellite data, which can limit the scalability and temporal analysis required for continuous urban climate monitoring.

2.3 Scalable Mapping with Sentinel-2

Addressing the need for scalable and frequently updated data, recent studies have increasingly adopted *Sentinel-2* imagery for forestry and urban green space applications. Notably, Freudenberg et al. [20] introduce a dedicated Sentinel-2 machine learning dataset specifically designed for tree species classification in Germany. Their work underscores the viability of using freely available, multi-spectral satellite data for large-scale vegetation mapping.

2.4 Multi-Temporal Analysis and Phenology

Furthermore, recent advancements highlight the critical role of multi-temporal analysis in vegetation mapping. Because the spectral signature of trees changes significantly throughout the year due to phenological cycles, relying on single-date imagery often leads to misclassifications. Grabska et al. [21] demonstrate that incorporating the entire Sentinel-2 time series—capturing leaf-on, leaf-off, and flowering periods—greatly enhances the ability of models to differentiate vegetation types. Integrating these temporal signals into deep learning architectures, such as through convolutional temporal attention networks, allows models to prioritize the most meaningful seasonal features for precise identification [15, 21].

2.5 Synthesis and Proposed Approach

Overall, while existing literature predominantly focuses either on VHR imagery for exact crown delineation or on traditional models for general species classification, our project bridges these areas. By leveraging the temporal resolution of Sentinel-2 data [20, 21] combined with a temporal attention *U-Net* architecture [15], this study aims to provide a scalable and robust solution for urban tree detection that does not rely on costly VHR imagery.

3 Data Sources

This project uses three main types of data: tree register records [8, 7, 9, 10, 11, 12], Sentinel-2 satellite imagery [22], and OpenStreetMap (OSM) urban features [23]. Together, these sources support the goal of mapping urban tree canopy from freely available data, which were collected, cleaned, and saved in reusable formats across four *Jupyter* notebooks [24, 25, 26, 27]. The data sources and their cleaning processes are described in the following subchapters.

3.1 Tree Register Data

Tree register datasets from six cities are included: Vienna [7], Linz [8], Innsbruck [9], Paris [10], Barcelona [11], and Hamburg [12]. The datasets were loaded in various formats, including *GeoJSON*, *JSON*, *CSV*, and *Excel*, and provide point-based information about known public trees. They are the main source of tree locations in this project to predict on Sentinel-2 satellite data.

Data preparation followed several steps: First, duplicates were checked across all datasets; none were found. Next, columns not needed for modelling were removed, while the following most useful variables were kept: geometry, species name, height, trunk circumference, crown diameter, and a city identifier. Coordinate systems were converted to WGS84 (EPSG:4326) and column names were standardised to make the datasets comparable across cities. Missing values were handled differently depending on the city. Innsbruck, Paris, and Barcelona contain no missing values in the retained columns after cleaning. For Hamburg Port data (hamburg-hpa), crown diameter and circumference contained missing values, and rather than dropping these rows, the missing values were imputed using the within-species median, with the overall column median used as a fallback. The same strategy was applied to Linz, where height, crown diameter, and circumference had a small share of missing values, around 4 percent of rows. Missing species names across all datasets were replaced with the placeholder value “unknown”.

For Vienna, the source data described missing entries in German as “nicht bekannt”, meaning “not known”, and “nicht definiert”, meaning “not defined”. These were also replaced with “unknown”. Additionally, young trees recorded with zero height, zero circumference, and zero crown diameter, labelled as “Jungbaumpflanzung”, were removed, as they represent newly planted trees with no measurable canopy. German species names were stripped from the species column. Vienna reports tree height not as a continuous value but as one of eight categorical height classes. For the exploratory data analysis part, midpoint values were suggested.

After cleaning, the register data were saved in *GeoPackage* and *Parquet* formats for efficient reuse. Barcelona was saved as Parquet, as it does not have a *GeoPandas* geometry column.

3.2 Sentinel-2 Satellite Imagery

For Sentinel-2 data collection, we first considered the Copernicus API [6], but then switched to *Google Earth Engine* [22] to be able to create monthly median composites from Sentinel-2 images filtered by cloud coverage. This makes the process more reproducible and reduces cloud-related problems compared with manual image selection in the previous project. Google Earth Engine also allowed us to restrict the data retrieval to the exact area used in the tree register data. To the six cities mentioned above, five additional cities—Prague, Budapest, Warsaw, Milan, and Athens—were added as new prediction targets for the final database. Their bounding boxes were obtained using the *geopy* library [28].

The project uses seasonal imagery for spring (April), summer (August), and autumn (November), chosen to capture different stages of vegetation development. For Paris and Prague, October was used instead of November, as no suitable November images were available. For Athens, July was used instead of August, as no images were available for that month. A cloud coverage threshold of 20 percent was applied; a stricter 10 percent threshold was tested first but did not produce usable images for several cities and months. From Sentinel-2, four spectral bands are kept: B02 (blue), B03 (green), B04 (red), and B08 (near-infrared), which are the only bands with 10-metre resolution [29]. In addition, three vegetation indices are computed: NDVI, EVI, and SAVI [30]. Only images from 2025 were used to keep the data consistent with the register sources, and the result is a multi-season spectral stack for each city, ready for training and prediction.

3.3 OpenStreetMap Urban Features

Tree register data cover publicly managed trees, so the locations of private trees are unknown. To provide additional spatial context for modelling, urban feature layers were collected from OpenStreetMap (OSM). These data come from *Geofabrik* downloads [23] in GeoPackage format. Regional files are available for Austria, Ile-de-France, Catalonia, and Hamburg. From each source, thematic layers were read: roads, buildings, land use, and water.

After loading and converting the layers to EPSG:4326 [31], they were clipped to the bounding boxes derived from the cleaned register data for each city. This keeps the register data, Sentinel-2 imagery, and OSM layers spatially aligned. At this stage, geometry validity and empty geometries were checked across all layers; no issues were found. No strong filtering of feature classes was applied at this step, in order to preserve flexibility for later modelling. Deeper cleaning is deferred to the exploratory data analysis phase. The cleaned layers were saved as GeoPackage and Parquet files for Vienna, Paris, Hamburg, and Barcelona in one notebook, and for Linz and Innsbruck in a second notebook, using the same Austria GeoPackage file, clipped to the respective city bounds.

3.4 Additional Viennese Context Data

Since OSM data is community-driven and can be spatially incomplete, and registers only cover public trees, additional data was collected. Since Vienna has particularly good and complete open data provision, three additional official datasets were collected specifically for Vienna to provide richer spatial context for modelling. To help the model learn the distinction between public and potentially private tree locations, the following official datasets were fetched:

RNK Land Use Data [32] provides official Vienna land-use polygons covering 30 land-use categories grouped into built-up area, green space, and transport. The top-level grouping was used, as it is simpler and more suitable for modelling. German category names were translated to English, and polygon centroids were converted to WGS84.

Public Green Space Data [33] contains official polygons of publicly accessible green areas in Vienna. The dataset distinguishes between sub-types such as forest, water, park, playground, dog zone, and sports areas. These sub-type columns were converted to boolean flags, making the dataset more informative for modelling.

The Vienna Multi-Purpose Map (Vector Data) [34] offers detailed urban surface and infrastructure polygons for Vienna, covering 24 feature types including buildings, roads, green spaces, forests, and water bodies. Duplicate rows, which represented only a small portion of the data, were removed, and rows with missing values were dropped. German feature type names were translated to English.

All three datasets were saved as Parquet files, ready for use in exploratory data analysis and modelling.

4 Methodology

For this tree detection task we implemented a *U-Net* regression architecture with a temporal attention mask which predicts the number of trees per pixel of *Sentinel-2* data where the prediction labels are derived from communal tree registers.

4.1 Data Processing

The first methodological step was fetching the different datasets on Sentinel-2 images, tree locations and OSM information. Figure 1 shows an overview of the Data Processing.

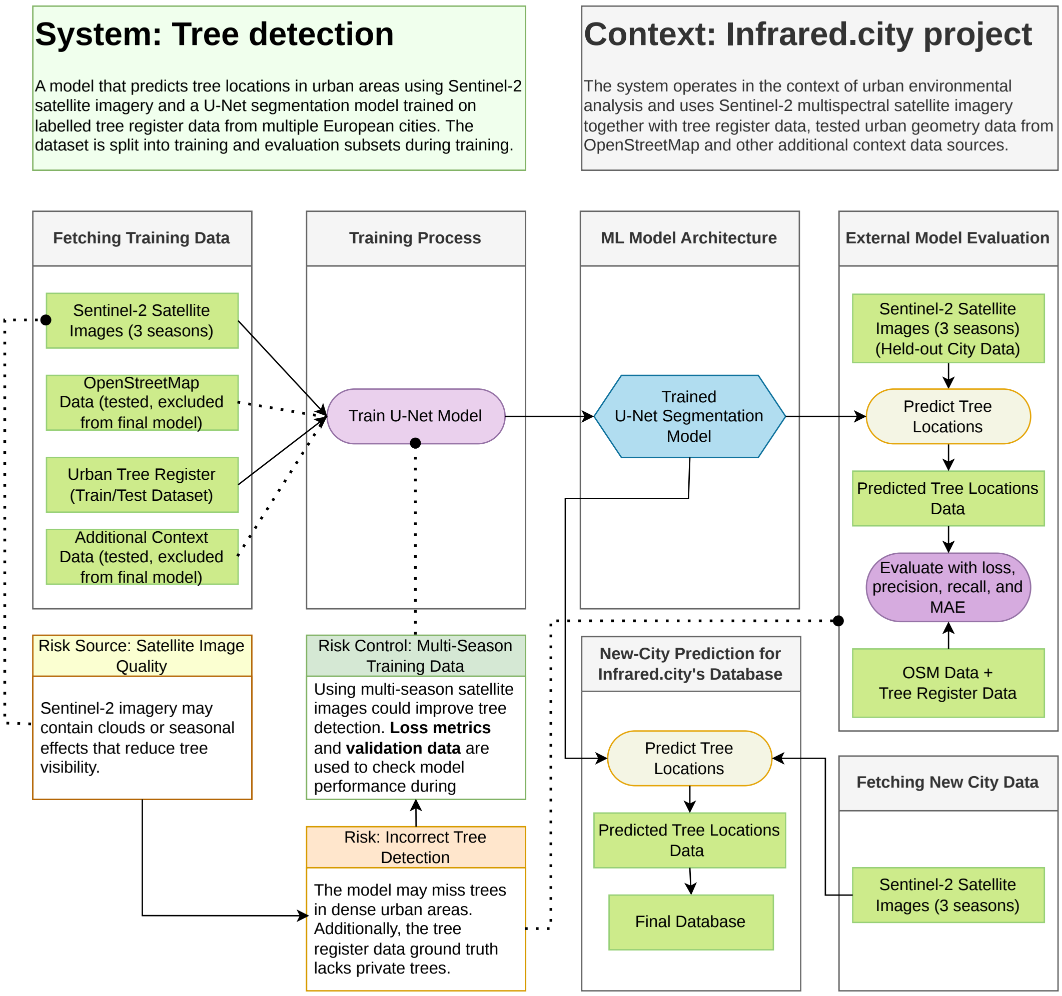


Figure 1: BEAM model of the proposed tree detection system.

4.2 Data Pipeline for ML

Through our EDA and some manual checking we were able to find some important properties about our data which influence the needed preprocessing pipeline and the model architecture. The most important aspects were that only around 4% of pixels contained trees in the ground truth and that most trees on private property were missing in our ground truth.

The first step in our preprocessing pipeline for the machine learning part of our process was to split the Sentinel-2 images into smaller patches which were easier to load into memory and to process for our model. To end up with more patches we also implemented stride, so patches were overlapping.

We then aligned the ground truth from the tree registers with the Sentinel-2 patches, labeled each patch with the corresponding month when the image was taken and normalized the values so it would be easier for the model to learn their patterns. As the data was already very sparse, we decided to drop patches with a low number of trees.

By manual checking we noticed that there was a significant number of trees missing in our ground truth which is due to the fact that trees on private property are not registered in the public tree registers. Our first attempt to solve this problem was to get data on what is private property, but this was more tedious than we expected because there is no complete available data on what is private property for any city.

We then noticed that the ground truth was normally very clustered, meaning that normally pixels containing trees in the ground truth were closer together and there were significant holes between them with many pixels containing zero trees. Which makes sense as private and public areas also are somewhat clustered. So, we switched to an approach where we would only use pixels for training where there were trees in a specified radius to avoid training on those large private areas falsely containing zero trees in the ground truth. Similar approaches already delivered promising results in former projects [35].

4.3 Vegetation Indices

Three spectral vegetation indices are derived from the Sentinel-2 surface-reflectance bands and used as additional input channels alongside the raw spectral bands. All three exploit the characteristic optical behaviour of healthy vegetation: strong absorption in the red part of the visible spectrum due to chlorophyll pigments, and strong reflection in the near-infrared (NIR) due to internal leaf-cell structure. In the formulas below, ρ_{Blue}, ρ_{Red}, and ρ_{NIR} denote surface-reflectance values in the range $[0, 1]$, corresponding to Sentinel-2 bands B02, B04, and B08 respectively.

Normalised Difference Vegetation Index (NDVI)

The Normalised Difference Vegetation Index [36] quantifies the contrast between red and near-infrared reflectance, normalised by their sum:

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}}} \tag{1}$$

NDVI takes values in $[-1, 1]$. Bare soil and built surfaces typically yield values between -0.1 and 0.2; sparse or stressed vegetation falls in the range 0.2–0.4; dense healthy canopy such as mature urban tree crowns produces values above 0.5. Water surfaces return negative values because NIR reflectance is lower than red reflectance over open water. Because the formula is a ratio of two quantities in the same physical units, NDVI is scale-invariant: it produces identical results whether computed from raw digital number (DN) values or from reflectance-calibrated values, which is why it is unaffected by the scaling correction required for EVI and SAVI (see below).

Enhanced Vegetation Index (EVI)

The Enhanced Vegetation Index [37] was developed to address two known limitations of NDVI: saturation under high-canopy conditions, where NDVI values compress into a narrow range even as canopy density continues to increase; and sensitivity to aerosol scattering in the atmosphere. EVI incorporates a blue-band correction term to reduce aerosol influences and a canopy-background adjustment in the denominator:

$$\mathrm{EVI} = G \cdot \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + C_1 \cdot \rho_{\mathrm{Red}} - C_2 \cdot \rho_{\mathrm{Blue}} + L} \tag{2}$$

The coefficients adopted in the MODIS EVI algorithm, and used in this project, are $G = 2.5$ (gain factor), $C_1 = 6$, $C_2 = 7.5$ (aerosol resistance weights), and $L = 1$ (canopy background adjustment). With these coefficients and reflectances in $[0, 1]$, EVI is bounded within $[-1, 1]$. The additive constant $L = 1$ in the denominator is critical: it prevents near-zero denominators and keeps the index bounded. When EVI is applied to raw DN values in the range 0–10,000, this constant is negligible relative to the band magnitudes, which causes EVI to lose its bounds—the scaling artefact identified during the exploratory data analysis and subsequently corrected by dividing all band values by 10,000 before computing the index.

Soil Adjusted Vegetation Index (SAVI)

The Soil Adjusted Vegetation Index [38] addresses the influence of exposed soil on vegetation index values. In urban environments with sparse canopy—street trees, recently planted areas, or autumn scenes where deciduous canopy has thinned—the underlying soil or hard surface contributes substantially to the pixel reflectance, causing NDVI to underestimate vegetation density. SAVI corrects for this by introducing a soil-brightness correction factor L into the NDVI denominator:

$$\mathrm{SAVI} = \frac{(\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}})\,(1 + L)}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}} + L} \tag{3}$$

The factor L is chosen based on vegetation density: $L = 0$ corresponds to very dense vegetation, where soil is not visible and SAVI reduces to NDVI; $L = 1$ corresponds to very sparse or absent vegetation with a dominant soil signal. The standard value $L = 0.5$ is used throughout this project, as Huete (1988) found it minimises soil brightness variations across a wide range of conditions without requiring additional calibration for different soil types. With $L = 0.5$ and reflectances in $[0, 1]$, SAVI is bounded within $[-1, 1]$. SAVI is expected to be most informative in autumn imagery, where leaf loss exposes underlying surfaces and NDVI becomes ambiguous.

4.4 Model

Former papers suggest that for image segmentation using satellite data the U-Net deep learning model is the most accurate for detecting objects, for example street detection [14] as well as crop field detection [15]. Furthermore, last year's *infrared.city* project also tried implementing a U-Net architecture but failed to yield promising results, also the data coaches deem it a well-fitting model.

The U-Net model is a special type of Convolutional Neural Network and consists of a contracting path and an expanding path.

First, the satellite image is put through an input convolution, so it has the correct structure for further processing. It is then passed into the contracting path to down-sample the data in the convolution and pooling layers, which are used to extract features from the original imagery. Our model uses four convolutional layers where during contraction for each layer the image size gets smaller due to pooling meanwhile evermore features are extracted from the images. The same process happens vice versa for the expansionary path. The expanding path consists of the same structure of convolution and pooling layers but this time used to up-sample the before contracted imagery [13].

As suggested in similar work this project will use skip connections between the expanding and contracting path as well as a temporal attention algorithm with multi-head attention to prioritize more meaningful images from the three months [15] and take advantage of the seasonality and changing vegetation of trees.

The outputs of the up-sampling block are then going through two separate final convolutional blocks where one head is supposed to predict the probability of a pixel containing any trees and the other head outputs the number of trees for each pixel. The two results are then multiplied to get the final prediction. This method helps overcome the problem that most pixels in the ground truth contain no trees and the issue that the labels are incomplete by enabling the model to learn uncertainty about presence in the probability head and about the abundance of trees in areas where they likely exist in the count head. The multiplication process suppresses false positive counts in areas with obviously no trees [39, 40].

During the training process the results of the probability head together with the ground truth are put through a weighted binary cross-entropy loss as we want results between 0 and 1 [41]. The results of the count head are first log transformed to ensure that outliers do not have too much of an impact on the gradients, then smooth L1 loss was used due to its robustness to labeling errors [42].

Figure 2 presents the BEAM representation of the proposed tree detection model.

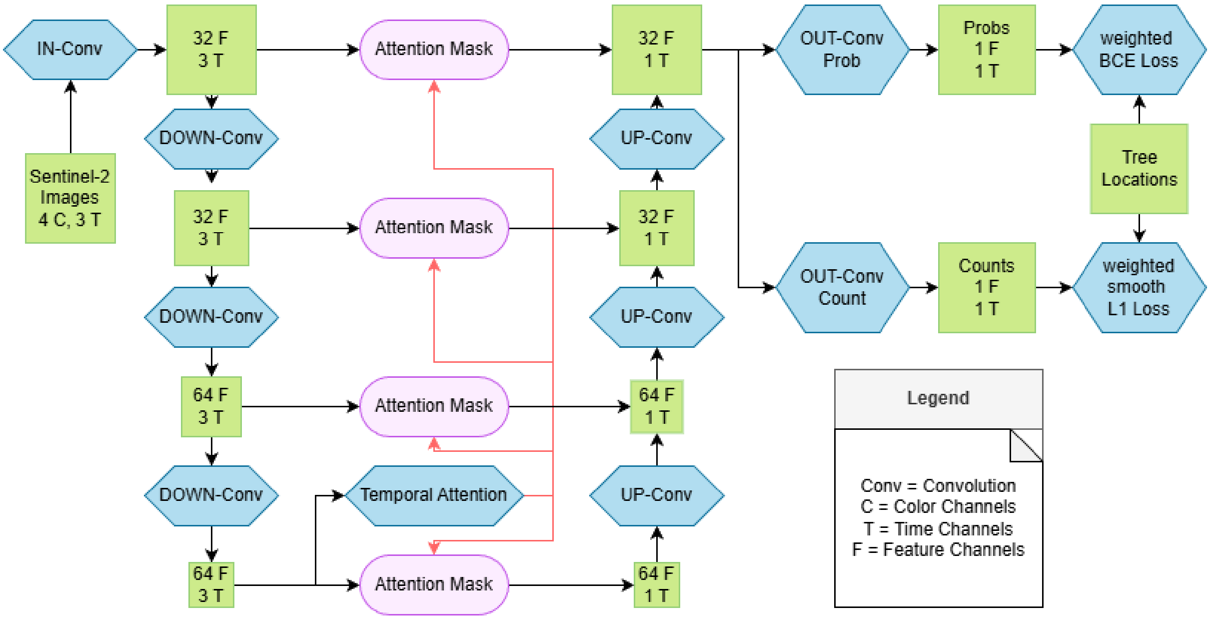


Figure 2: BEAM model of the proposed tree detection model

5 Implementation

This chapter describes the workflow and hyperparameters used for data preprocessing for machine learning as well as the final model training, testing and predicting.

5.1 Tools

This model was implemented using the programming language *Python* and the deep learning library *PyTorch*. To store the data in an easily machine-readable format suited for machine learning tasks, the library *NumPy* was used, and the data was stored as NumPy arrays. To read the .geojson files and to convert the final predictions back into this file format, the library *GeoPandas* was used. For data preprocessing, the libraries *Rasterio* and *Pyproj* were used.

5.2 Data Pipeline for ML

In the preprocessing pipeline for the U-Net model the following steps and hyperparameters were used. All of them were decided upon after testing different alternatives. The decisions were mostly based on the metrics described in the next section *Model Training, Testing and Predictions*.

For the final patches we decided to have a height and width of 256 pixels and a stride of 128 pixels. We also only used the five color bands red, green, blue, near infrared and NDVI as the other two bands did not improve model performance meaningfully but increased the needed training time significantly. These patches were then created for all three months used which were normally April, August and November but for some cities October was used instead of November as data was too cloudy. The patches for the months were then stacked on top of each other to end up with tensors of shape $N \times T = 3 \times C = 5 \times H = 256 \times W = 256$, where N is the amount of patches in the city, T the number of temporal observations (months), C the number of spectral channels, and H and W the spatial dimensions (height and width) respectively.

The data from the tree registers was then geographically aligned with the Sentinel-2 patches to get the ground truth containing the amount of trees per pixel. Patches with less than ten trees were then dropped to decrease sparsity. Overall, we ended up with around 1400 usable patches.

The final step was to implement the algorithm to determine which parts of each patch should be used for training the model. We ended up using all pixels which had a tree in a five pixel radius as this was a good approximation of what area is correctly labeled in the ground truth and what area is not.

The data was then split into a train, validation and test sample. The training and validation data consisted of six cities, where each city's data was split into 15% validation data and 85% training data. For testing we wanted to have a city where not even parts were used for the training process as this would also be the case for cities we wanted to do predictions on. We chose Barcelona, which is our southernmost city, as a city for testing as this would also allow us to see if the model could generalize on cities which were a bit different than our training data.

The data we used for our final predictions was preprocessed in a similar way, of course excluding the non-existent tree register and without defining a usable area as we want predictions on the whole batch.

5.3 Model Training, Testing and Predictions

The model was built according to the description in the section *Methodology*, although the depth of the model is limited due to computational restraints. All hyperparameters listed here were chosen as to optimize the final loss, the MAE, recall and precision.

At the bottleneck, where the feature space is the richest and the attention algorithm is deployed, the U-Net had 64 feature channels, before down-sampling 32 and after up-sampling 32. The attention mechanism uses 4 attention heads, the QKV vectors have 8 dimensions, the embedding vectors 32. At the bottleneck the attention algorithm combines the three images for each month into one image using the scaled dot-product of the learned query vectors and adds positional encodings. This step ensures that the most important features from each month are included in the final image.

As described in the section *Methodology*, the model uses two heads where one predicts the probability of a pixel containing trees and the other the number of trees per pixel. The probability head uses BCE loss where pixels containing trees are weighted twice as much as pixels containing no trees. This is due to the fact that we can say that if our ground truth contains a tree this is correct but we cannot be as sure if the ground truth contains no tree. The results of the count head are put through a softplus activation function to ensure they are positive. Once again, the positive pixels are weighted more than the negative, here by a factor of ten. Of course both losses are only calculated on pixels which were deemed usable in the data preprocessing. The losses are then summed where the loss of the probability head is weighted five times more than the loss of the count head.

The used learning rate is 0.0001 and training ran for five epochs with a batch size of 4. To reduce the likelihood of overfitting all images were randomly rotated and flipped for each epoch. To evaluate the overall model the final loss was used. To evaluate the results of the probability head its own loss, precision and recall were calculated, the threshold was 0.5. For the results of the count head the MAE for each pixel was calculated. The model was then tested on the data from the unseen city.

6 Results

This chapter presents the results of the exploratory data analysis conducted across the three data sources: the tree registers, the *Sentinel-2* satellite imagery, and the *OpenStreetMap* (OSM) urban context layers, as well as the extended analysis of two additional training cities, Innsbruck and Linz. The goals of the analysis were to assess data quality, validate feature distributions, and identify the indicators most likely to be informative for the *U-Net* model.

6.1 Tree Register Analysis

In this section, we analyze the structural and spatial properties of the collected tree register datasets. We evaluate the varying tree densities across cities, discuss label reliability and temporal misalignments, outline necessary data corrections (with a particular focus on Vienna), and examine the species composition to understand the underlying canopy variations.

6.1.1 Dataset Overview and Tree Density

The cleaned tree register data contain 226,998 trees for Vienna, 217,454 for Paris, 233,505 for Hamburg, and 43,367 for Barcelona. Comparing these raw counts directly is misleading because the bounding boxes scraped for each city differ substantially in area. Normalising by the official administrative area of each city (as illustrated in Figure 3) yields a more informative comparison: Paris registers approximately 2,063 trees per km^2 of city area, far exceeding the other three cities, while Hamburg, whose scraped bounding box covers nearly 4,000 km^2, has the lowest density at 309 trees per km^2.

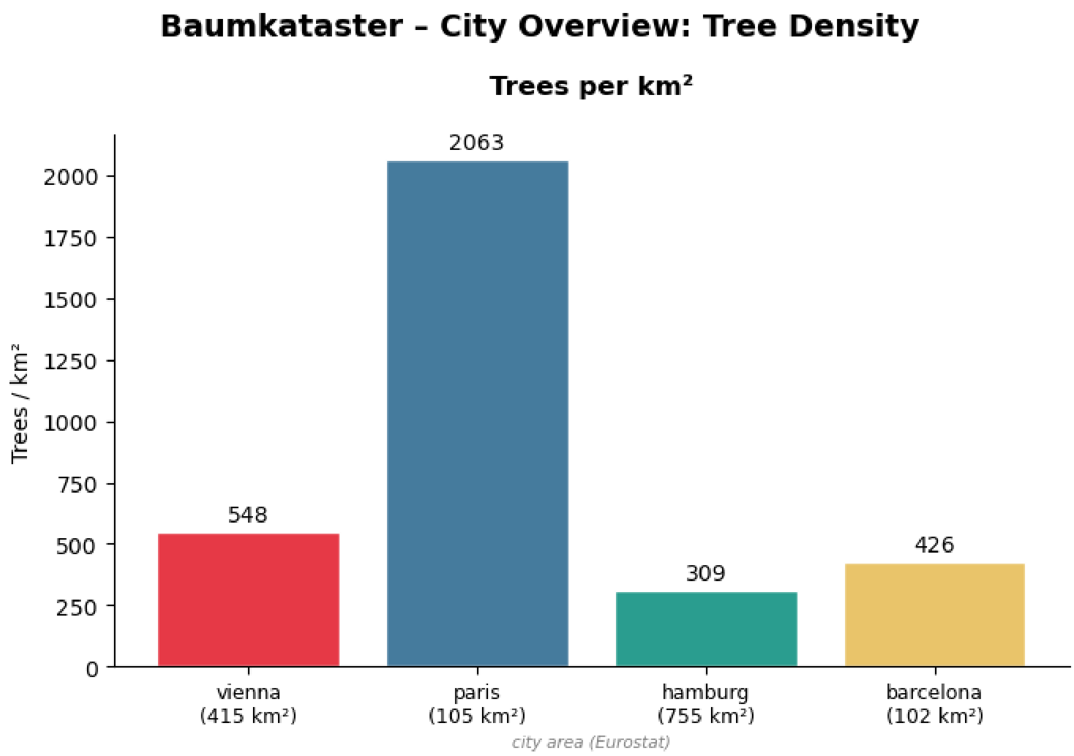


Figure 3: Tree density per km² of administrative area per city.

6.1.2 Label Reliability: Temporal Misalignment, Private Property, and Removed Trees

Three systematic biases in the tree register labels are relevant to any modelling interpretation of performance metrics.

Temporal misalignment. The tree register datasets are snapshots rather than continuously updated registers. Vienna and Hamburg are updated roughly annually; the Paris dataset was last substantially revised around 2023; the Barcelona dataset dates from approximately 2017–2019. The Sentinel-2 composites used in this project are 2025 median mosaics, placing the tree register labels two to six years behind the imagery. A tree planted after the last tree register update that is already visible in 2025 will carry no positive label; a tree that has since been removed will retain one. Both cases inject noise into the supervision signal and set a practical ceiling on the model's achievable recall and precision when measured against the tree register.

Private-property trees. Municipal tree registers register only publicly owned or publicly accessible trees. Trees on private land, such as courtyard trees, garden trees, trees on institutional grounds are almost entirely absent. Barcelona is the most extreme case, partly explaining its comparatively low count despite a per-km^2 density otherwise comparable to Vienna. For the model this creates a systematic false-negative pressure during training: the satellite detects green NDVI signal from unregistered private canopy, but those pixels carry no positive label and are treated as background. The task should therefore be framed as detecting *canopy* rather than detecting *registered trees*, with precision and recall measured against the tree register as an operationally useful proxy rather than a complete ground truth.

Removed trees. The inverse problem: a tree register entry surviving after a tree has been felled or died introduces a complementary false-positive pressure. The NDVI signal at that location resembles built or bare surface, but the label is positive. This effect is geographically scattered and smaller in magnitude than the private-property gap, but it cannot be resolved without a tree register snapshot contemporaneous with the imagery.

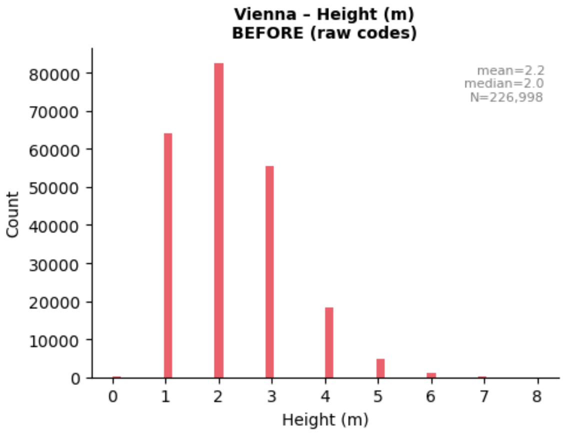
6.1.3 Vienna Data Corrections

Two encoding issues specific to Vienna were identified and corrected before any further analysis. First, the `height` column stores ordinal category codes 0–8 representing height intervals from <5 m to >35 m, not continuous metres. Second, the `crown_diameter` column uses the same ordinal scheme linked to crown-diameter intervals. Both were decoded to interval midpoints, as detailed in Table 1. Additionally, 415 Vienna circumference values of zero were found to encode unknown measurements and were replaced with NaN.

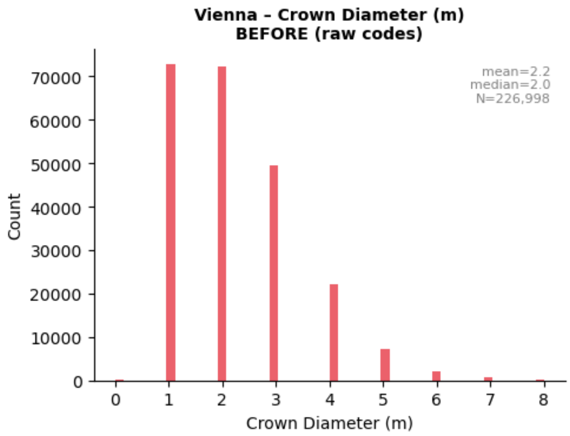
After decoding, the Vienna height distribution has a median of 8.0 m and a mean of 8.9 m, closely matching Paris (median 8.0 m, mean 9.0 m), validating the decoding scheme against an independent source. The corrected Hamburg trunk circumference mean (121.6 cm) exceeds Vienna (98.0 cm) and Paris (82.3 cm), consistent with Hamburg's dominance by large-crowned *Quercus robur* and *Tilia × europaea*. Figure 4 illustrates the size distributions across cities before and after these corrections were applied.

Table 1: Vienna ordinal code decoding scheme applied to `height` and `crown_diameter`.

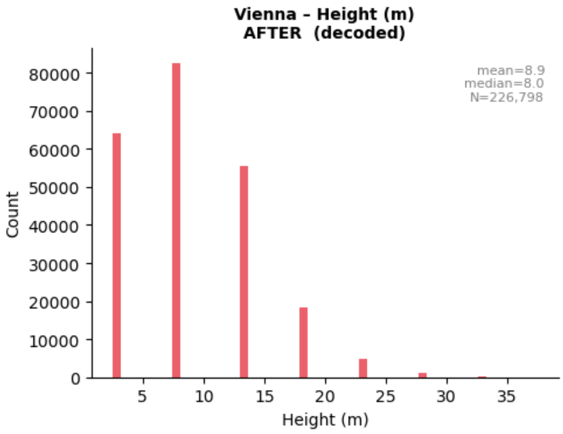
Code	Height class	Midpoint (m)	Crown class	Midpoint (m)
0	nicht bekannt	NaN	nicht bekannt	NaN
1	0–5 m	2.5	0–3 m	1.5
2	6–10 m	8.0	4–6 m	5.0
3	11–15 m	13.0	7–9 m	8.0
4	16–20 m	18.0	10–12 m	11.0
5	21–25 m	23.0	13–15 m	14.0
6	26–30 m	28.0	16–18 m	17.0
7	31–35 m	33.0	19–21 m	20.0
8	>35 m	37.5	>21 m	22.5



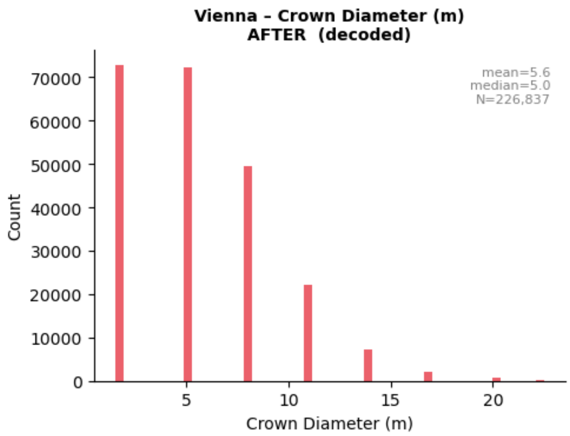
(a) Height (raw codes)



(b) crown diameter, before correction (raw codes).



(c) tree height, after correction (decoded to metres).



(d) crown diameter, after correction (decoded to metres).

Figure 4: Vienna size distributions before and after correction.

6.1.4 Species Composition

Species diversity is high across all cities: 670 unique species in Vienna, 996 in Paris, 354 in Hamburg, and 330 in Barcelona, with 86 species shared across all four cities, predominantly from the *Abies* and *Acer* genera. Vienna is dominated by *Acer platanoides* (Norway maple, 19,244 trees) and *Aesculus hippocastanum* (horse chestnut, 11,661 trees), both classic central-European street species. Paris is dominated by *Platanus* × *hispanica* (London plane, 38,113 trees). Hamburg is dominated by *Quercus robur* (English oak, 42,965 trees) and *Tilia* × *europaea* (European lime, 33,852 trees). Barcelona shows a distinct Mediterranean composition including subtropical species such as *Tipuana tipu* and *Pinus pinea*. Species identity is not a direct model input, but species diversity informs expectations about canopy shape and size variation across cities.

6.1.5 Spatial Distribution and Label Imbalance

Figures 5, 6 and 7 reveals that tree locations are distributed unevenly within each city. Raw patch sampling would therefore oversample dense clusters and underrepresent sparser districts, potentially producing a model that performs well on dense parkland but fails on isolated street trees. The patch-sampling strategy must account for this imbalance, for instance through stratified sampling or spatially weighted loss functions.

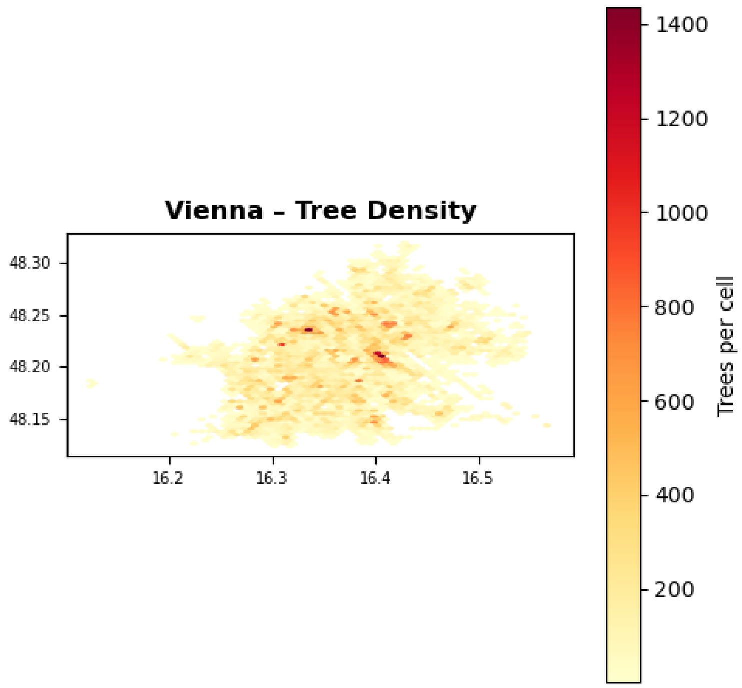


Figure 5: Spatial distribution of labelled tree locations in Vienna.

6.2 Sentinel-2 Analysis

All 12 city-season Sentinel-2 composites (four cities × three seasons) were successfully loaded and validated. The seven available channels are B02 (blue), B03 (green), B04 (red), B08 (NIR), NDVI, EVI, and SAVI.

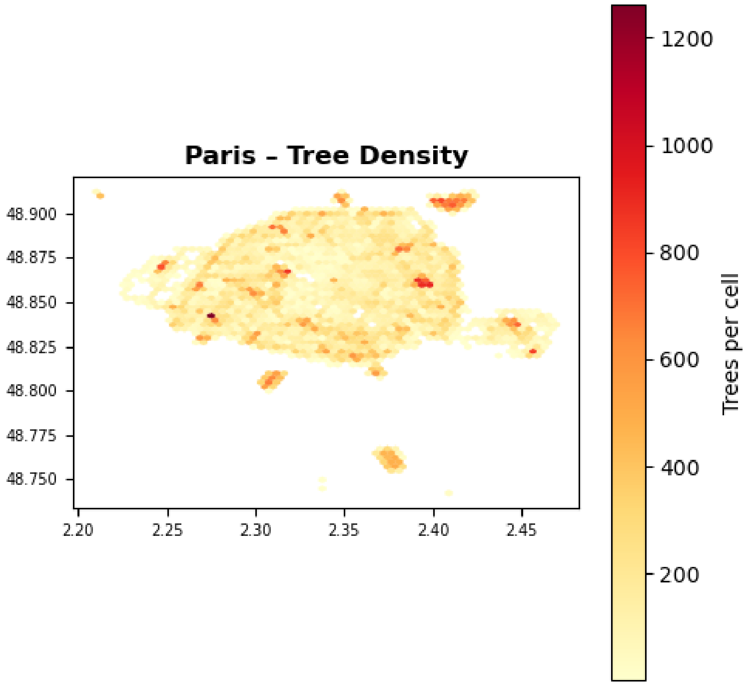


Figure 6: Spatial distribution of labelled tree locations in Paris.

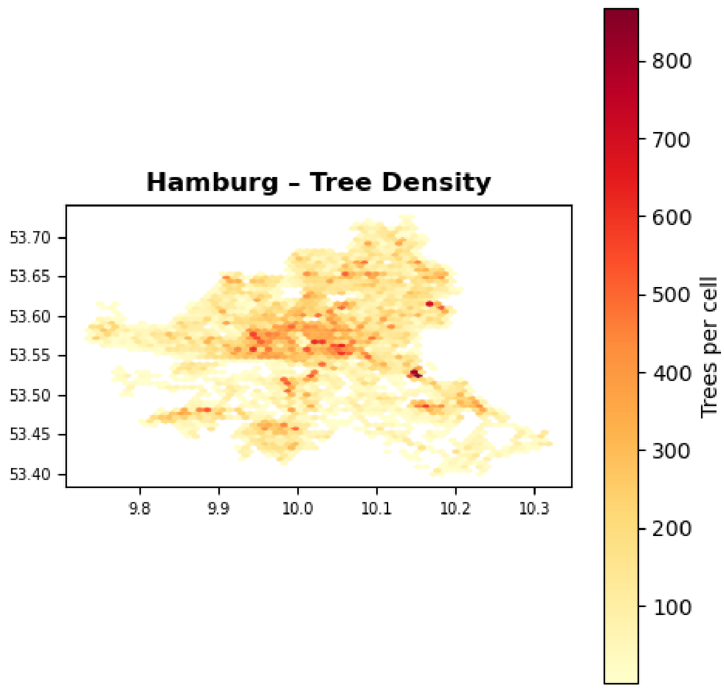
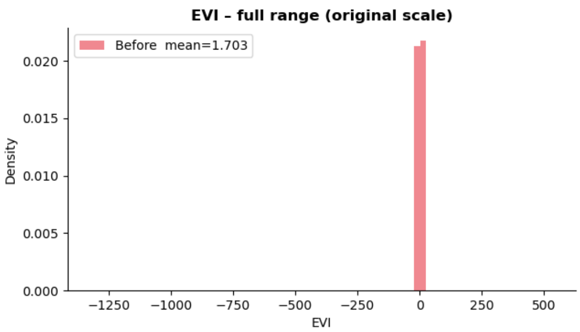


Figure 7: Spatial distribution of labelled tree locations in Hamburg.

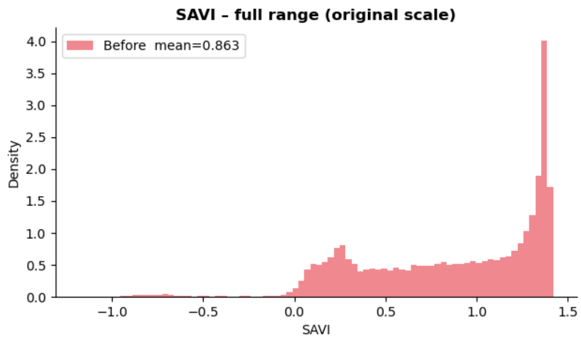
6.2.1 EVI and SAVI Scaling Correction

A systematic quality issue was identified: EVI and SAVI were computed from raw Sentinel-2 digital number (DN) values in the range $\approx$ 0–10,000 rather than from surface-reflectance values in $[0, 1]$. Because both formulas contain an additive constant in the denominator (EVI: $+1$; SAVI: $+0.5$), this constant is overwhelmed when band values are on the order of 10^3, causing EVI to substantially exceed its theoretical bound of $+1.0$. NDVI is unaffected because it is a pure ratio and is therefore scale-invariant.

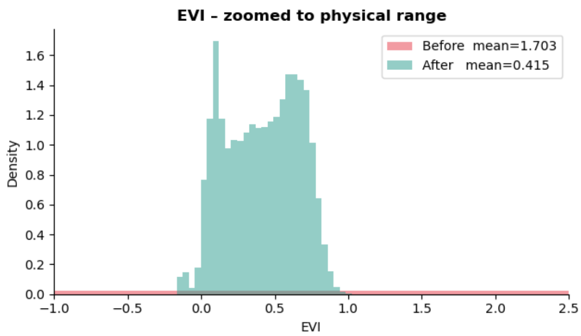
The correction was applied by dividing all raw band values by 10,000 before recomputing EVI and SAVI. After correction, EVI values lie within $[-1, 1]$ across all cities and seasons, as demonstrated by the comparison in Figure 8, confirming the fix. Corrected TIF stacks were saved to `data_after_eda/sentinel data/`.



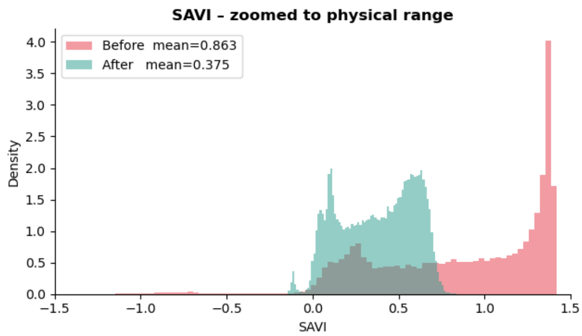
(a) EVI — full range (original scale)



(b) SAVI — full range (original scale)



(c) EVI — zoomed to the physical range



(d) SAVI — zoomed to the physical range

Figure 8: EVI and SAVI scaling correction.

6.2.2 NDVI Seasonal Patterns

Table 2 details the mean NDVI values per city and season. The seasonal pattern is city-specific rather than uniform, and a visual summary of these patterns is provided in Figure 9.

Vienna and Hamburg show their highest NDVI in summer (0.577 and 0.418 respectively), consistent with temperate deciduous canopies reaching full leaf-out in August. Barcelona is near-flat across all three seasons (0.281–0.287), reflecting its Mediterranean, evergreen-dominated species composition and mild seasonal amplitude. Paris is the notable exception: its autumn NDVI (0.476) exceeds both its spring (0.371) and summer (0.396) values, an anomaly that requires a contextual explanation.

Table 2: Mean NDVI by city and season. The Paris autumn value corresponds to October imagery, as no November composite was available.

City	Spring (Apr)	Summer (Aug)	Autumn (Oct/Nov)
Vienna	0.563	0.577	0.457
Paris	0.371	0.396	0.476
Hamburg	0.295	0.418	0.347
Barcelona	0.281	0.287	0.287

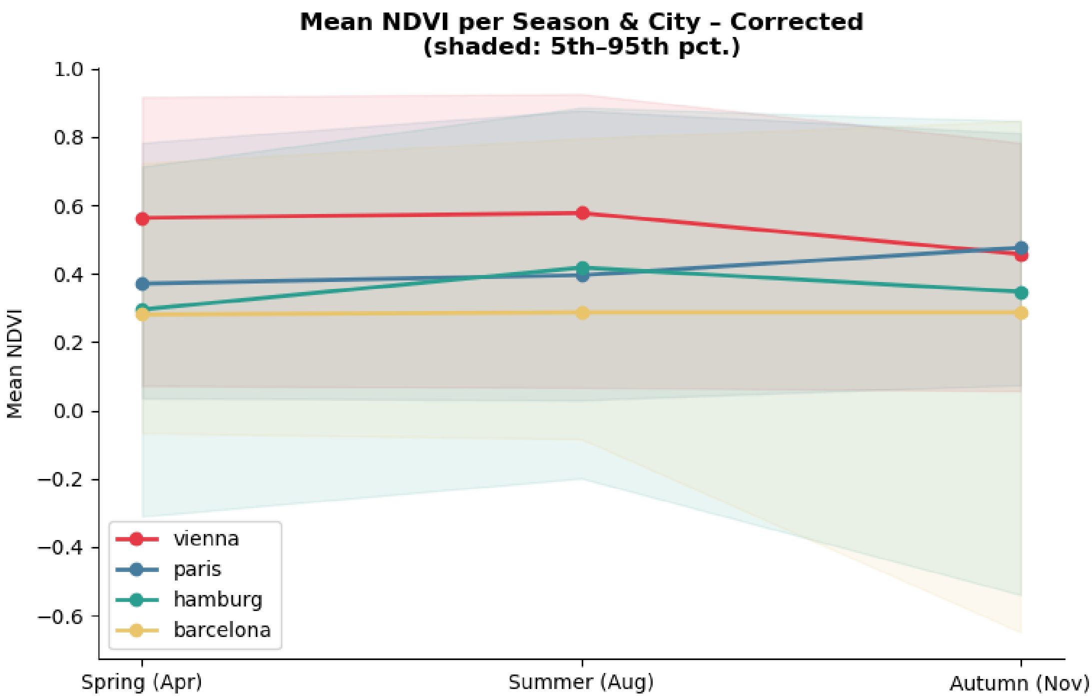


Figure 9: Mean NDVI per season and city.

6.2.3 Paris NDVI and the 2024 Olympics Urban Reforestation

The Paris autumn NDVI value (0.476) exceeds both its spring (0.371) and summer (0.396) means, inverting the temperate-deciduous pattern seen in Vienna and Hamburg. It is tempting to read this as a signature of the city's large-scale greening programme: under a 2020 commitment, Paris set out to plant 170,000 trees between 2020 and 2026, with roughly 8,900 of them concentrated in and around the Olympic Village in Seine-Saint-Denis for the 2024 Summer Games (The Local, 2020; Monocle, 2026). Because the Sentinel-2 composites used here are 2025 mosaics, they postdate both the Games and much of this effort.

On closer inspection, however, attributing the autumn peak to reforestation is difficult to defend. First, the planting is poorly matched to what a coarse vegetation index can resolve. Only about 63,500 of the 170,000 target had been planted by mid-2023 (Euronews Green, 2023), so a large share of the trees visible to the 2025 imagery are one to three growing seasons old; newly planted street trees have crowns far smaller than the 10 m Sentinel-2 pixel and contribute negligibly to pixel-level NDVI in their first years. Second, a single city-wide seasonal mean is too aggregate an instrument to isolate a spatially concentrated intervention: even if the Seine-Saint-Denis plantings raised NDVI locally, that signal would be diluted across the full administrative area and would not preferentially inflate the *autumn* value relative to summer.

A simpler explanation is already present in the data. As noted in Table 2, the Paris "autumn" composite is built from October imagery because no cloud-free November mosaic was available, whereas the other cities use November. October in Paris falls before full leaf senescence, so a green October composite sits systematically above a true late-autumn one. The apparent autumn > summer inversion is therefore most parsimoniously a compositing artefact rather than a phenological or anthropogenic one. A depressed August value under summer heat or drought stress could compound the effect, but this cannot be confirmed from the present data and is not asserted here.

This reframing matters for how the reforestation enters the project. The seasonal NDVI curve is derived from reflectance and is independent of the tree register, so the planting programme does not "explain" the NDVI anomaly. Its concrete consequence is instead one of label quality: the Paris tree register was last substantially revised around 2023 and predates most Olympic-era planting, so the 2025 imagery contains canopy, particularly in the northern arrondissements, that carries no positive label. Paris is therefore the city with the most severe label-to-image temporal misalignment in the corpus, and evaluation metrics computed over Parisian patches in these districts should be read with that undercount in mind.

6.2.4 False-Color Composites

Figure 10 displays false-color composites (NIR → red, red → green, green → blue) for all four cities across all three seasons, rendered from corrected TIF stacks. In this encoding, healthy vegetation appears bright red, built surfaces grey-cyan, and water near-black. The composites confirm that the Sentinel-2 data are correctly georeferenced and that vegetation is clearly separable from urban surfaces across all cities and seasons. Seasonal changes in canopy density are visible for Vienna and Hamburg between summer and autumn. Barcelona's canopy remains consistent across all three composites with its near-flat NDVI.

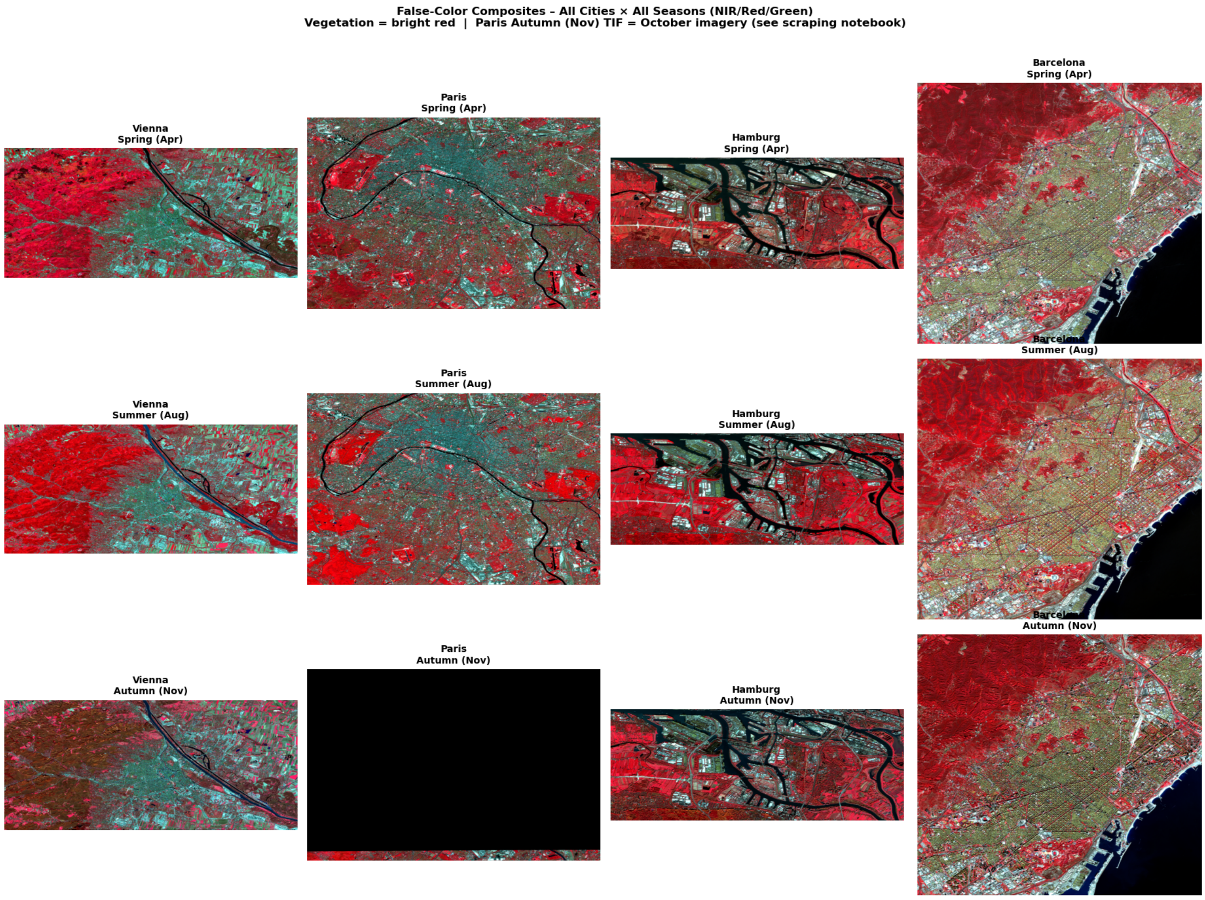


Figure 10: False-color composites (NIR/Red/Green) for all four cities.

6.3 OSM Analysis

This section assesses the OpenStreetMap (OSM) data used to provide spatial context for the model. As a community-driven project, OSM exhibits varying coverage and semantic consistency across different cities. Here, we examine the distribution of road types and land use zones to determine their suitability as negative-label masks or contextual covariates.

6.3.1 OSM as a Community Project: Cross-City Coverage and Semantic Consistency

OpenStreetMap is a volunteer-contributed dataset, and its coverage and the *semantics* of its tags vary significantly across cities. This has direct consequences for any analysis that treats OSM labels as ground truth for vegetation or land cover.

Vienna benefits from one of the most active national OSM communities in Europe. The Austrian contributor base maintains fine-grained `park`, `forest`, and `grass` tags that are generally reliable and up to date.

Hamburg presents a documented gap: the road network is densely mapped (161,378 features), but the landuse layer is substantially sparser than expected given the city's comparable tree register count, with only 24,184 road features in the landuse extract. This reflects genuine mapping gaps in residential green space rather than a scraping or clipping issue.

Paris has strong OSM coverage in the central arrondissements but patchy coverage in the outer banlieues. Critically, the Seine-Saint-Denis districts that received the bulk of the Olympic-era tree plantings are likely under-mapped in the landuse layer, compounding the label-temporal-misalignment problem discussed in Section 6.2.3.

Barcelona has reasonable OSM coverage for public parks and major green zones, but private green space—which is substantial given Barcelona's dense, enclosed *illa* block structure—is almost never tagged.

Beyond coverage gaps, a deeper semantic problem exists: the tags `park`, `garden`, `grass`, and `forest` do not encode equivalent vegetation densities across cities. A Viennese *Stadtpark* is a formal, heavily tree-canopied park with high NDVI; a `park` tag in Barcelona may refer to a small, partly paved plaza with a few ornamental trees. Treating these as a binary green/not-green mask in the cross-dataset analysis of Section 6.1.1 conflates very different vegetation structures. For model training, OSM landuse is therefore best used as a *contextual covariate* that provides spatial priors about likely versus unlikely tree locations, not as ground truth for vegetation presence.

6.3.2 Road Type Distribution

Figure 11 breaks down the road-type composition per city. Footway, service, and residential classes dominate in Barcelona, Paris, and Vienna. Hamburg is the exception, where service roads outnumber footways. Only wider classes (primary, secondary, motorway) are suitable as reliable tree-free negative-label candidates for mask generation. Narrower classes such as path, steps, and track frequently run through parks alongside trees and must not be used as hard negative masks.

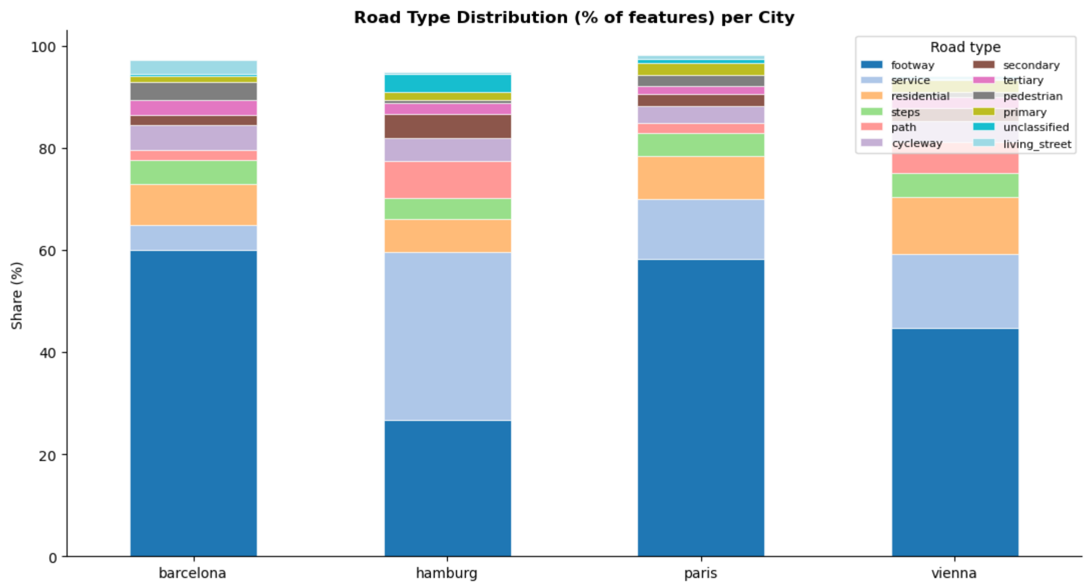
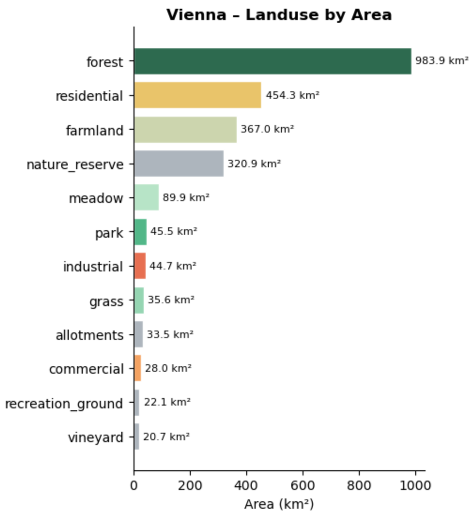


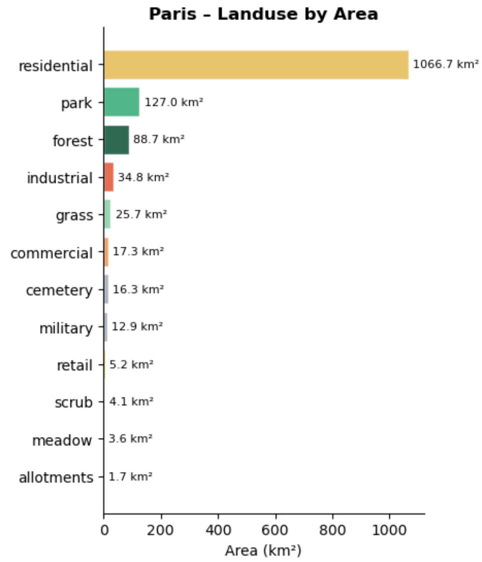
Figure 11: Road-type distribution per city as a percentage of total road features.

6.3.3 Land Use Distribution

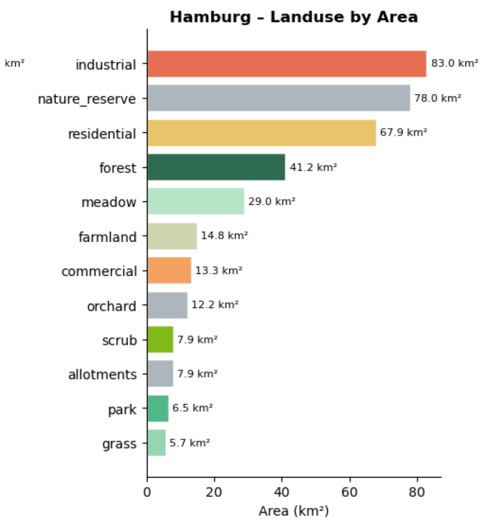
Figure 12 illustrates the land use distribution by area across the analyzed cities. Vienna and Paris contain large mapped green-area polygons; Hamburg's landuse layer is notably sparser, consistent with the mapping gap described above. Building footprints are spatially reliable for all four cities and will be used as hard negative-label zones during training mask generation, since no trees are expected on rooftop surfaces.



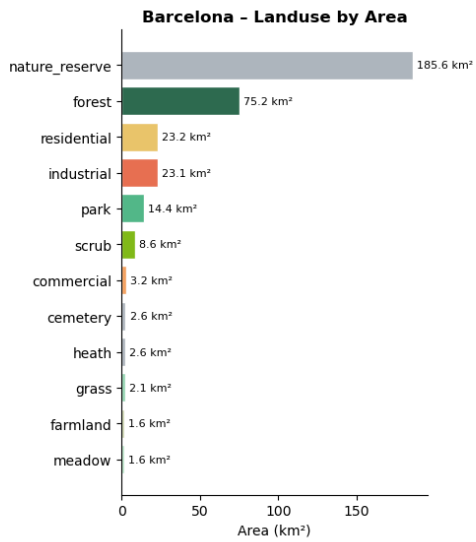
(a) Vienna



(b) Paris



(c) Hamburg



(d) Barcelona

Figure 12: Land use distribution by area per city.

6.4 Extended Training Corpus: Innsbruck and Linz

Two additional Austrian cities, Innsbruck and Linz, were added to expand the training corpus with urban environments of different morphological character and with a stronger Austrian context for generalisation testing.

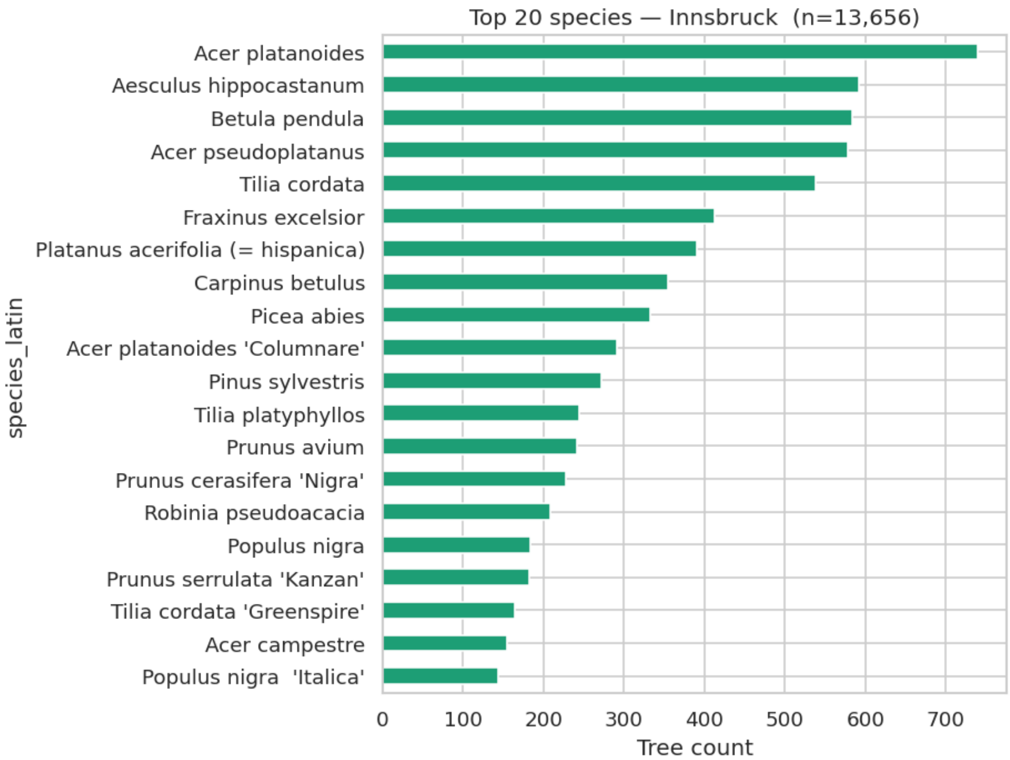
6.4.1 Tree Registers: Innsbruck and Linz

The Innsbruck tree register contains 13,656 trees described by longitude, latitude, species, and city identifier only; no morphological attributes are available. Linz provides a richer dataset of 26,866 trees including continuous height (m), crown diameter (m), circumference (cm), and a binary `type_of_tree` flag distinguishing deciduous (`L`, 24,053 trees, 89.5%) from coniferous (`N`, 2,813 trees, 10.5%). Table 3 summarises the key metrics of these two datasets.

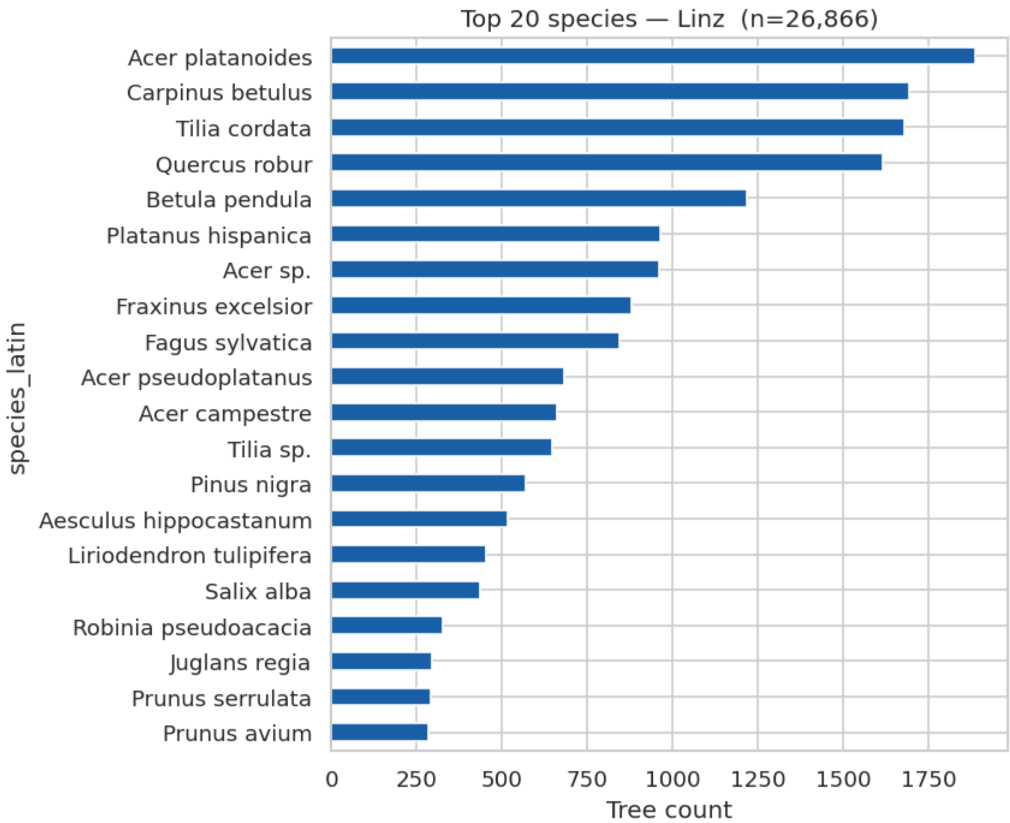
Table 3: Short summary for the Innsbruck and Linz tree register datasets.

City	**Trees**	**Species**	**Top species**
Innsbruck	13,656	304	*A. platanoides* (5.4%)
Linz	26,866	374	*A. platanoides* (7.0%)

Species diversity is consistent with the primary training cities, and the dominant species across both—*Acer platanoides* (Norway maple)—is also the leading species in Vienna, Hamburg, and Barcelona, confirming a coherent central-European temperate flora. 117 species are shared between Innsbruck and Linz (20.9% of the combined union of 561 species). Figure 13 shows the species distributions and long-tail analysis for both cities.



(a) Innsbruck



(b) Linz

Figure 13: Species distribution and long-tail analysis for Innsbruck and Linz.

6.4.2 Morphological Features and Data Quality: Linz

Figure 14 presents the distributions of trunk circumference versus height for Linz, together with species-level boxplots. The distributions are realistic and follow expected biological ranges for urban trees in a temperate climate.

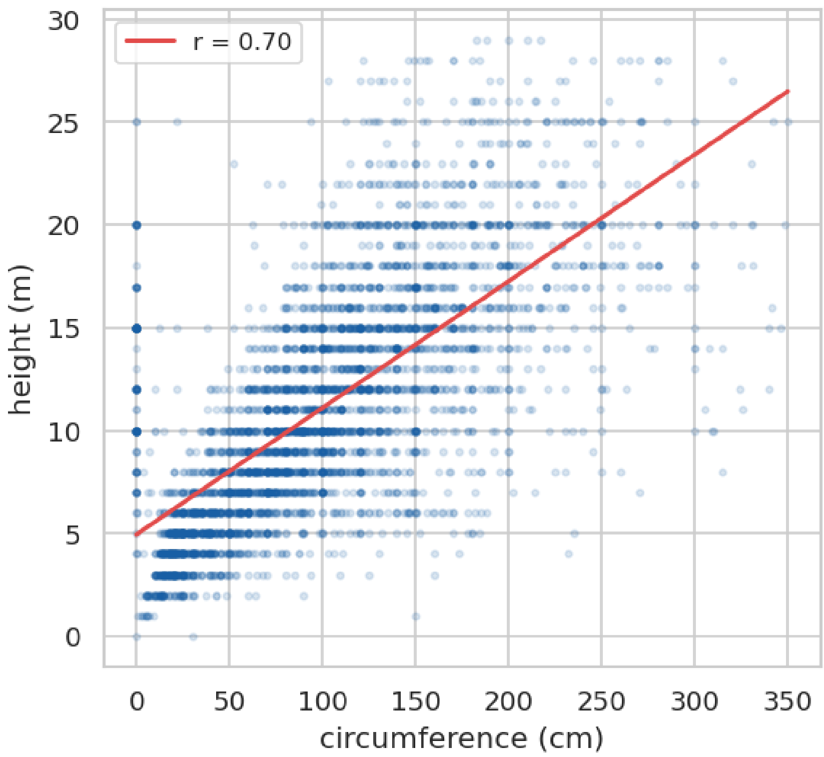


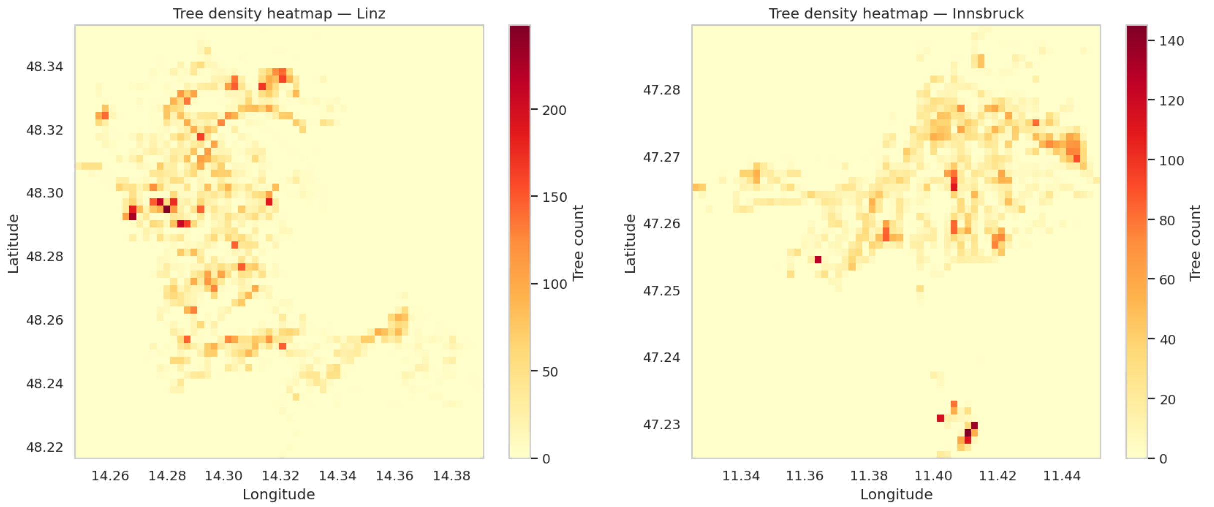
Figure 14: Trunk circumference vs. height. Morphological feature distributions for Linz.

IQR-based outlier inspection identifies a small number of likely data-entry errors at the top 0.1% of each column: one *Fagus sylvatica* record lists a height of 151 m (physically impossible; probable unit confusion) and one *Salix alba* record lists a circumference of 11,120 cm (likely entered in millimetres rather than centimetres). These records should be capped or excluded before model training.

Coniferous trees have a higher median height (13.0 m vs. 10.0 m for deciduous) and larger trunk circumference (110 cm vs. 90 cm), while deciduous trees have slightly wider crowns. These biologically plausible patterns validate data quality and suggest that the `type_of_tree` flag could serve as a useful auxiliary label if the model is extended to species-type classification.

6.4.3 Spatial Distribution: Innsbruck and Linz

Figure 15 visualizes the spatial distribution of trees in Innsbruck and Linz via density heatmaps. Both cities exhibit the characteristic non-uniform clustering seen in the primary training cities: trees are concentrated in boulevard networks and larger parks, with sparser coverage in industrial and newer residential zones. The combined corpus overview (further supported by the spatial coverage in Figure 16 and median polygon areas in Figure 17) confirms that adding Innsbruck and Linz increases the total labelled tree count by 40,522 trees and extends the geographic and morphological range of the training data.



(a) Linz (b) Innsbruck

Figure 15: Tree density heatmaps for Linz and Innsbruck.

6.4.4 OSM: Innsbruck and Linz

OSM data for both cities were loaded across roads, buildings, landuse, and water layers. Linz has 57,702 road features and 62,006 building footprints; Innsbruck has 22,064 and 24,337 respectively, proportional to their populations. Figure 16 outlines the feature-class distributions (spatial coverage) for both cities, while Figure 17 compares the median polygon areas by group. Both cities benefit from active Austrian OSM contributor communities and show comparably complete coverage to Vienna. Building footprints are spatially reliable for negative-label masking. The same cross-city semantic caveats about green-space tag interpretation apply, although the risk is lower in the Austrian context where OSM norms are more consistent.

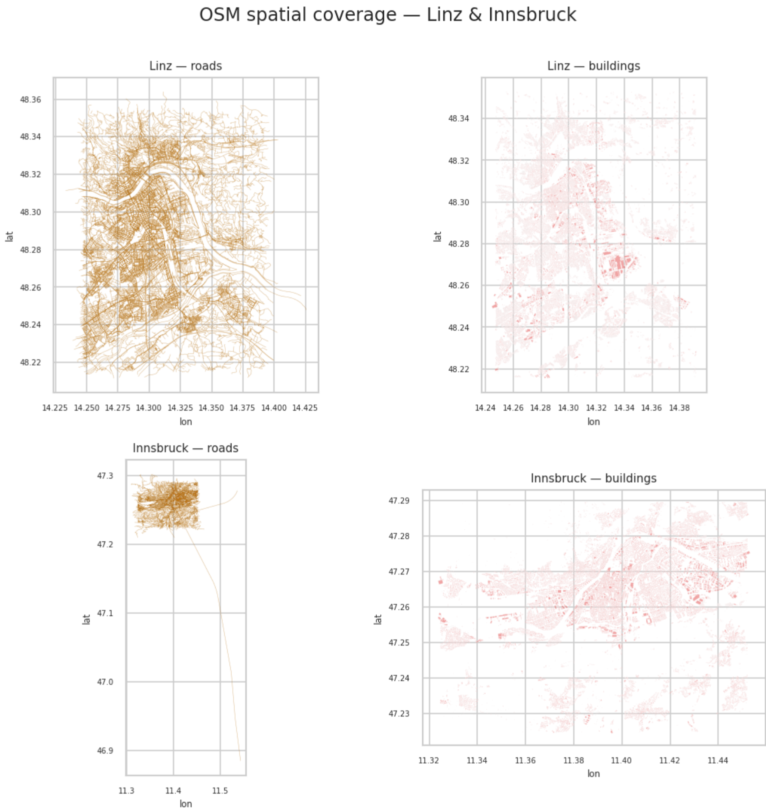
6.4.5 Sentinel-2: Innsbruck and Linz

Sentinel-2 composites were successfully retrieved for both cities across three seasons. Table 4 details the mean NDVI values for these additional cities by season.

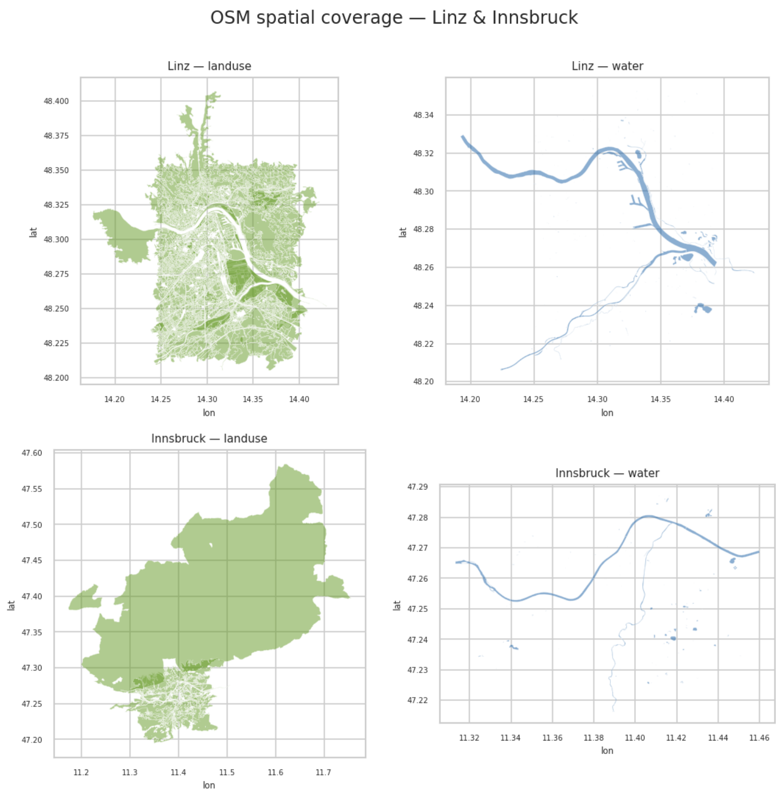
Table 4: Mean NDVI for Innsbruck and Linz by season (corrected TIFs, 2025 composites).

City	Spring (Apr)	Summer (Aug)	Autumn (Nov)	Δ Sp$\to$Su	Δ Su$\to$Au
Linz	0.555	0.568	0.509	+0.013	−0.059
Innsbruck	0.536	0.642	0.286	+0.106	−0.356

Linz exhibits a stable, high NDVI across all three seasons, suggesting a predominantly mixed-canopy character with dense urban green coverage consistent with an Austrian mid-sized city. Innsbruck shows a markedly different pattern: summer NDVI peaks at 0.642, higher than any of the primary training cities in summer, while autumn NDVI falls sharply to 0.286, a drop of −0.356 from summer. This pronounced phenological signal reflects Innsbruck's Alpine geography: the city is surrounded by deciduous valley-floor vegetation and steep forested slopes that lose their canopy rapidly in October–November. This steep gradient makes Innsbruck a particularly informative training example for the



(a) Roads and Buildings



(b) Landuse and Water

Figure 16: Spatial coverage Linz and Innsbruck.

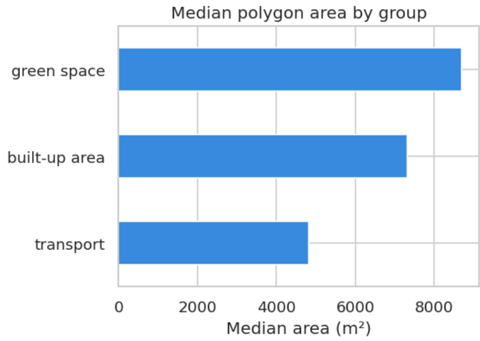


Figure 17: Median polygon area by group.

temporal attention mechanism, which must learn to weight the less-informative autumn timestep down relative to spring and summer.

Figure 18 shows the respective false-color composites across all three seasons. Healthy vegetation appears bright red. The strong seasonal change in Innsbruck is clearly visible in the autumn composite, where the valley vegetation and surrounding slopes substantially degreen.

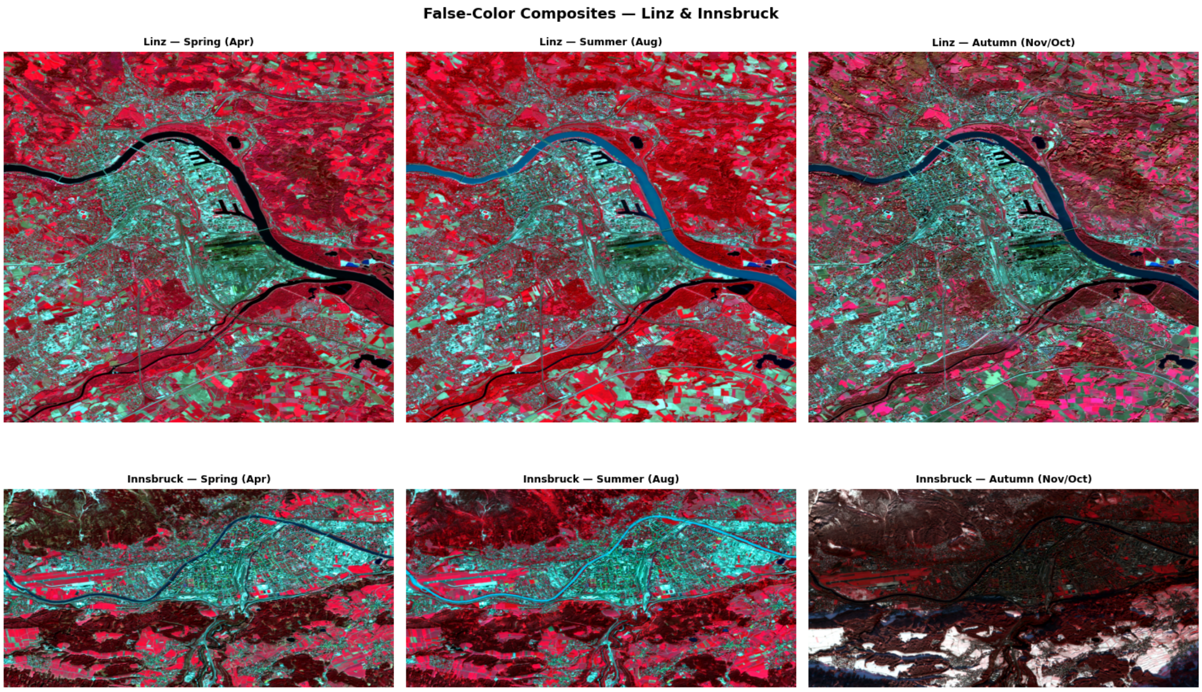


Figure 18: False-color composites (NIR/Red/Green) for Linz (top) and Innsbruck(bottom)

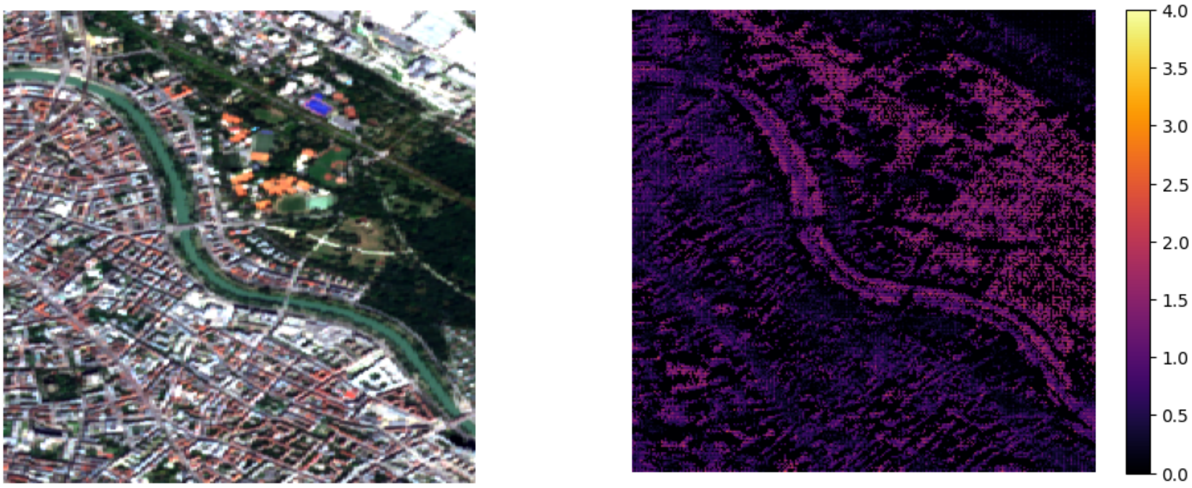
All six Innsbruck and Linz TIFs passed data-quality checks with more than 95% usable pixels. A single EVI artefact was found in the Innsbruck November file (0.76% of pixels out of bounds, mean value -1.04×10^8, caused by near-zero denominators in haze pixels over the Inn river). This was resolved by recomputing EVI with a denominator guard and masking affected pixels to NaN; corrected files were saved alongside the primary-city stack in `data_after_eda/sentinel data/`.

6.5 Model Results

At first sight, the model overall achieves poor results with precision around 30% and recall around 40% and MAE of a bit higher than 1.1. Most of this can be attributed to the model having difficulties detecting single trees at a very low resolution where each pixel is 10 m wide. Also, the imperfect ground truth with missing trees on private property makes it hard for the model to achieve good results. Other factors like trees being cut down during the year or small clouds which are difficult to detect for the model may also have an impact on model performance but are very negligible.

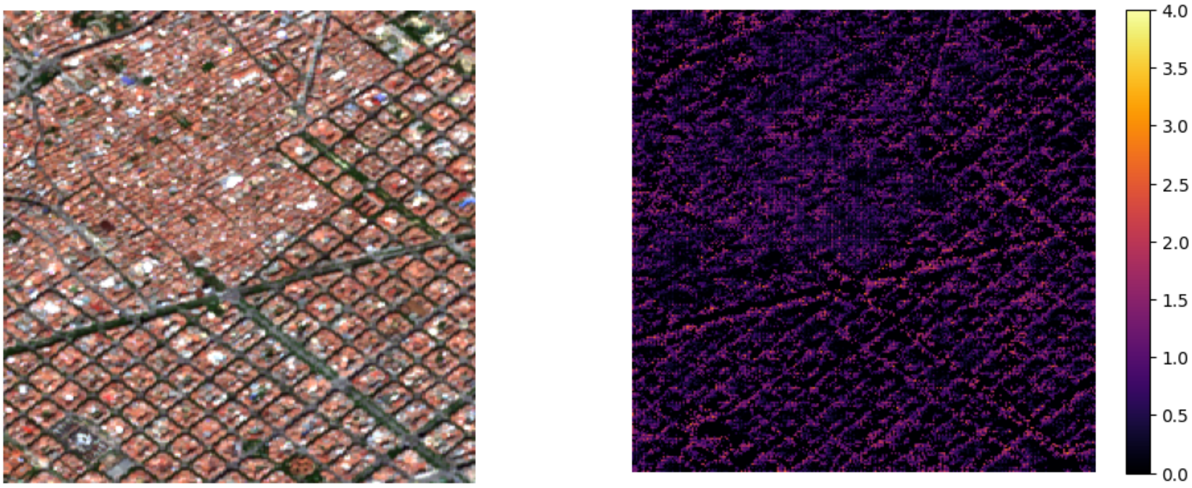
Better results can be seen when checking the actual predicted trees of the model manually, either by tediously checking for each tree on *Google Earth* or similar sites or by comparing the input image to the output image with the expected trees. So here are some representative examples both from Barcelona which was test data and from those parts of Vienna which were used for validation. Figure 19 and Figure 20 directly compare the RGB satellite imagery with the corresponding model predictions for Vienna and Barcelona, respectively.

In those pictures we can see some very interesting phenomena. Firstly, that the model predicts a lot more trees than are in the ground truth which is a good thing, as we know that a lot of trees are not labeled. Furthermore, it also detects trees in areas where no trees are in the ground truth, so the model learns how a tree looks and can implement this knowledge also on wrongly labeled pixels.



(a) RGB image (b) Model predictions

Figure 19: Comparison of RGB imagery and model predictions for Vienna (August 2025).



(a) RGB image - Barcelona (b) Model predictions - Barcelona

Figure 20: Comparison of RGB imagery and model predictions for Barcelona (August 2025).

7 Discussion

In this project, our main goal was to build a working machine learning model that can find trees in cities. We used *Sentinel-2* satellite images and open tree register data from the cities. During our work and the exploratory data analysis (EDA), we found out some interesting things about the data and also ran into some problems, especially because we did not have perfect ground truth data.

7.1 Ground Truth

When you first look at our evaluation metrics, the precision is only around 30%. Normally, this looks like a really bad model. But we have to understand why this happens. Our main dataset for the ground truth is the tree register. This dataset is from the city government, which means it only has trees that are on public streets or in public parks. Trees in private gardens or backyards are just missing.

Because of this, our ground truth is not complete. When our model finds a real tree in a private garden, the evaluation script says it is a "false positive" and our score goes down. But when we looked at the predictions by eye (for example in Vienna and Barcelona) and compared them with *Google Maps*, we saw that the model is actually right. The *U-Net* learned how a tree looks like from the satellite data and can find trees even if they are not in the city data. So, the 30% precision is more a problem of the missing data, and not because our model is bad.

7.2 Different Seasons and NDVI

We thought from the beginning that using different seasons is a good idea to find trees. The data showed us that this is true, but it also surprised us. In remote sensing, people often say that the NDVI is highest in summer. But our data showed different patterns for each city. Vienna and Hamburg had the peak in summer, like expected. But in Paris, the highest NDVI was in autumn, and for Barcelona the curve was almost flat all year.

This shows exactly why our temporal attention mechanism in the U-Net is so important. If we only used one image from summer, the model would not work well for cities like Barcelona where it is too hot and dry in August. With the attention layers, the model can choose itself which month is the most important one to look at.

7.3 Parks vs. Street Trees

We also used *OpenStreetMap* (OSM) to get more context and saw big differences between the cities. In Vienna and Paris, a lot of the labeled trees (around 50% and 40%) are inside green spaces like parks or forests. But in Hamburg, only 2.6% are in parks, and almost all of them are street trees.

This was a big problem for training. If a model just learns that "green polygon = trees", it will miss all the street trees in Hamburg. But our model still managed to find single trees in areas with a lot of buildings. This means it really learned the shape and color of the trees and did not just copy the green areas from OSM.

7.4 Data Mistakes and Problems

When you work with real open data, there are always problems. One big mistake we made was with the EVI and SAVI index calculations. The pixel values we got from Sentinel-2 are big numbers, and we forgot to divide them by 10,000 to get normal reflectance values between 0 and 1. Because of this, our calculated EVI values were over 1.0, which is physically wrong. So we had to drop EVI and SAVI for now and only used NDVI for the training.

Also, the different city datasets were hard to mix. For example, Vienna has the tree height saved as categories (classes 1 to 8), but Paris has the real height in meters. This made it very hard to build one big dataset for everything.

7.5 Why we used the Dual-Head U-Net

In the end, our data has a lot of pixels with zero trees. If you just use a normal model, it will often just predict zero everywhere because that is the easiest way to get a low loss. That is why we decided to use a dual-head U-Net.

One head of the model just predicts if there is a tree or not (probability), and the second head predicts how many trees there are (count). We also masked the training data so the model only learns from areas near known trees. Multiplying the outputs of both heads helped a lot to stop the model from predicting trees inside houses or streets. This architecture was a very good idea to handle the noisy and incomplete ground truth.

8 Future Work

Even though our *U-Net* model already works quite well for finding trees, there are still several areas we can improve. Because of time and computational constraints, we had to keep some things simple. Here are the most important ideas on what to do next.

8.1 Putting Trees and Temperature Together

Right now, we only predict where trees are. But other people in the *infrared.city* project are predicting the land surface temperature (LST) in the city. Trees make the city cooler because they give shade and water evaporates from their leaves. In the future, we should feed our tree maps directly into the temperature models. If the temperature model knows exactly where the trees are, it can simulate the urban heat islands much better.

8.2 Using More Colors (Bands) from Sentinel-2

For our model, we only used four standard bands from the *Sentinel-2* satellite: red, green, blue, and near-infrared. We did this to reduce our computational effort while handling the data during training. But Sentinel-2 actually has many more channels, like the "Red-Edge" and "SWIR" (Short-Wave Infrared) bands. Other researchers like Freudenberg et al. [20] showed that these extra bands are very good to see if a tree is healthy or what kind of tree it is. If we use all bands in the future, our model will probably make fewer mistakes and maybe even tell us if it is a coniferous or a deciduous tree.

8.3 Dealing with Clouds (Using Radar Data)

The biggest problem with Sentinel-2 is that it is an optical sensor. If there are clouds, we cannot see the trees. For example, in Paris we had to use October pictures because November was too cloudy. To fix this, we could use *Sentinel-1* data in the future [43]. Sentinel-1 uses synthetic aperture radar (SAR), which can just look through the clouds. The radar signal also bounces off the tree trunks and branches. If we mix the optical pictures with the radar data, we can always map the trees, even if the weather is bad.

8.4 Testing other Climates

Right now, our model only knows cities in the middle of Europe with a temperate climate. We have data for Barcelona, but we did not use it for training yet because the file formats were inconsistent and some columns were missing. In the future, someone has to clean the Barcelona data so we can use it. Trees in the south, like *Pinus pinea*, look very different from our northern trees, and their leaves do not get as green in the summer. We need to train the model on different climates so it works everywhere in the world.

9 Conclusion

In this project, our primary goal was to build a machine learning model capable of detecting trees in urban environments. This is crucial for our partner *infrared.city*, as they need precise tree locations to simulate urban temperatures during the summer. We utilized free *Sentinel-2* satellite imagery and open tree register data from various cities to train a specialized *U-Net* model.

The most important result is that our model actually performs very well, even if the formal evaluation metrics initially suggested otherwise. Because the public tree registers only include public trees, our precision score was negatively impacted when the model correctly identified trees in private gardens. However, visual inspections confirmed that the model's predictions are highly accurate. It successfully learned the visual characteristics of a tree from the satellite data and can find them everywhere, not just on public streets.

Additionally, incorporating imagery from three different seasons proved to be a very effective strategy. Our data analysis revealed that the NDVI patterns change significantly depending on the city. Thanks to the temporal attention layers, the model was able to learn these patterns and autonomously determine which month is the most informative for each specific location.

The dual-head U-Net architecture was the right choice for this specific problem. Because our urban maps contain many pixels with zero trees, a standard model would often default to predicting zero everywhere to minimize the loss. By splitting the model into two parts—one predicting the probability of a tree's presence, and the other counting the abundance—we achieved much better results and significantly reduced false predictions on buildings or streets.

Naturally, there is still work to be done in the future. We need to further test the model on cities with different climates, such as Barcelona. Ultimately, however, we successfully built a working, scalable prototype. In the future, the tree maps generated by our model can be integrated directly into temperature models to pinpoint exactly where trees provide shade and help cool down the city.

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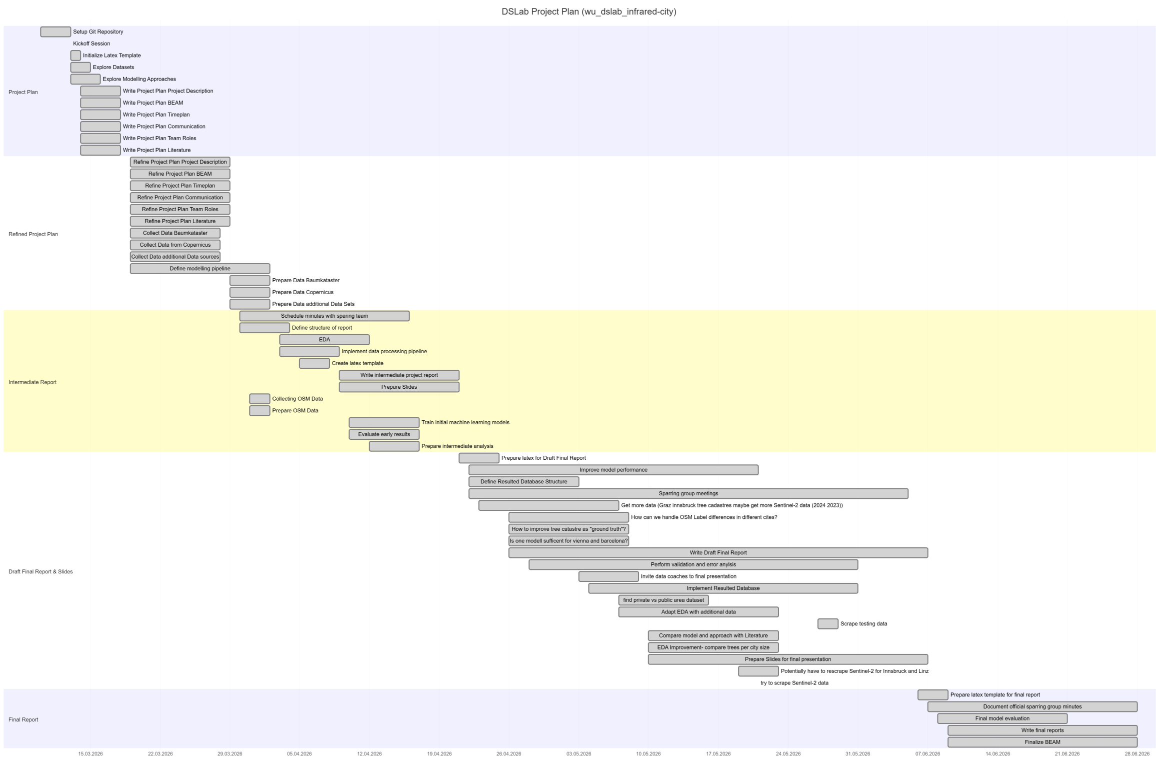
A GANTT

Exported GANTT Diagram to show progress over tickets in project.
Export on 28. April 2026



B Project Contribution

This sections desribes who contributed what to the project.

Moritz Hörmansdorfer

Moritz Hörmansdorfer was heavily involved in the conception and development of the modeling and data processing pipelines. His core responsibilities included exploring modeling approaches, defining the data pipeline, as well as training and evaluating the initial machine learning models. Another main focus of his work was critically examining the data foundation, for example, by investigating the tree cadastre as a "ground truth" and considering whether a single model is sufficient for different climate zones (like Vienna and Barcelona). Furthermore, he was responsible for defining and implementing the final database structure. Like all team members, he actively participated in general project planning, report writing, and preparing presentations.

Nina Salnikow

Nina Salnikow acted as the primary data manager within the project. She led the exploration, collection, and preparation of various datasets, including tree cadastres, Copernicus data, OpenStreetMap (OSM), and other potential sources for different cities (e.g., Graz, Innsbruck). She handled complex scraping tasks, evaluated the feasibility of utilizing Sentinel-2 data, and explored ways to differentiate between private and public areas. Compiling the testing data also fell under her purview. In addition, she was deeply involved in creating the BEAM (Business Event Analysis and Modelling) project plan and took charge of documenting the official sparring group minutes.

Dominik Oberhumer

Dominik Oberhumer played a central role in the technical project administration and the structuring of the reports. He set up the Git repository and provided the LaTeX templates for all necessary documentation. In terms of project planning, he focused on communication, role distribution, and drafting the schedule, which included the integration of GANTT charts. Furthermore, he was primarily responsible for the formal structuring of the intermediate and final reports and led the comparison of the team's model with existing literature (e.g., Freudenberg et al.). The organizational tasks regarding the sparring groups and inviting the data coaches to the final presentation were also managed by him.

Sara Klasová

Sara Klasová focused on literature research, Exploratory Data Analysis (EDA), and systematic model validation. She was responsible for drafting the project plan sections concerning the schedule and relevant literature. A major focus of her work was conducting and continuously adapting the EDA, such as integrating new data from Graz and Innsbruck or comparing tree populations based on city size. She also dealt extensively with methodological challenges, like handling different OSM labels across various cities. In the later phases of the project, her core competencies were applied to improving overall model performance, performing validation, and conducting detailed error analysis.

C Project Tickets

This sections shows the title of all tickets the persons worked on.

Moritz Hörmansdorfer

- Kickoff Session
- Explore Modelling Approaches
- Write Project Plan Project Description
- Refine Project Plan Project Description
- Define modelling pipeline
- Implement data processing pipeline
- Train initial machine learning models
- Evaluate early results
- Prepare intermediate analysis
- Write intermediate project report
- Prepare Slides for intermediate presentation
- How to improve tree catastre as ”ground truth”: historically trees? trees on private ground?
- Is one modell sufficent for vienna and barcelona? or define multiple models per ”climate-zone”
- Improve model performance
- Define Resulted Database Structure
- Implement Resulted Database
- Sparring group meetings
- Write Draft Final Report
- Prepare Slides for final presentation
- Final model evaluation
- Write final reports
- finalize BEAM

Nina Salnikow

- Kickoff Session
- Explore Datasets
- Write Project Plan BEAM
- Refine Project Plan BEAM
- Collect Data Baumkataster
- Collect Data from Copernicus
- Collecting OSM Data
- Collect Data additional Data sources?
- Prepare Data Baumkataster
- Prepare Data Copernicus
- Prepare OSM Data
- Prepare Data additional Data Sets?
- Define modelling pipeline
- Implement data processing pipeline
- Prepare intermediate analysis
- Write intermediate project report
- Prepare Slides for intermediate presentation
- Get more data (Graz, innsbruck tree cadastres, maybe get more Sentinel-2 data (2024, 2023))
- try to scrape ['B2', 'B3', 'B4', 'B5', 'B6', 'B7', 'B8', 'B8A', 'B11', 'B12'] sentinel-2 data -¿ goes beyond our scope
- Potentially have to rescrape Sentinel-2 for Innsbruck and Linz -¿ need EDA results
- find private vs public area dataset
- Scrape testing data
- Define Resulted Database Structure
- Sparring group meetings
- Write Draft Final Report
- Prepare Slides for final presentation
- Final model evaluation
- Write final reports
- Document official sparring group minutes
- finalize BEAM

Dominik Oberhumer

- Kickoff Session
- Setup Git Repository
- Initialize Latex Template
- Write Project Plan Communication
- Write Project Plan Team Roles
- Refine Project Plan Timeplan (GANTT code inside!!)
- Refine Project Plan Communication
- Refine Project Plan Team Roles
- Prepare intermediate analysis
- Schedule minutes with sparing team
- Define structure of report (which chapters? table of content)
- Create latex template for intermediate report
- Write intermediate project report
- Prepare Slides for intermediate presentation
- Compare model and approach with Literature (reimplement and compare result?): M. Freudenberg, S. Schnell, and P. Magdon, “A sentinel-2 machine learning dataset for tree species classification in germany,”
- Define Resulted Database Structure
- Sparring group meetings
- Prepare latex for Draft Final Report
- Write Draft Final Report
- Prepare Slides for final presentation
- Invite data coaches to final presentation
- Final model evaluation
- Prepare latex template for final report
- Write final reports

Sara Klasová

- Kickoff Session
- Explore Modelling Approaches
- Write Project Plan Timeplan
- Write Project Plan Literature
- Refine Project Plan Literature
- Define modelling pipeline
- EDA
- Implement data processing pipeline
- Prepare intermediate analysis
- Write intermediate project report
- Prepare Slides for intermediate presentation
- Adapt EDA with additional data (graz, innsbruck, sentinal-2 averages)
- How can we handle OSM Label differences in different cites [it is just a community project]? (in EDA and model)
- EDA Improvement: compare trees per city size, instead of overall trees
- Improve model performance
- Perform validation and error anylsis
- Define Resulted Database Structure
- Sparring group meetings
- Write Draft Final Report
- Prepare Slides for final presentation
- Final model evaluation
- Write final reports